

SINOPSIS KURSUS / COURSE SYNOPSIS

Jabatan Biologi / Department of Biology

BGY3000* Prinsip Biologi / *Principles of Biology* 4(3+1)

Prasyarat : Tiada / None

Kursus ini merangkumi teori sel dan organisasi sel dalam pengawalan homeostasis sebagai asas kepada semua proses fisiologi. Konsep pewarisan genetik dan hubungan ekologi dalam kemandirian organisma juga dibincangkan.

This course covers the cell theory and cell organisation in regulating homeostasis as basis of all physiology processes. Concept of genetic heredity and ecological interaction in organisms survival are also discussed.

BGY3002 Biologi Sel dan Molekul / *Cell and Molecular Biology* 3(2+1)

Prasyarat : Tiada / None

Kursus ini merangkumi ciri molekul yang membentuk sel dan asas kimianya, kaitan antara struktur komponen sel dan fungsinya, serta interaksi antara sel dengan persekitarannya. Aliran tenaga di dalam sel, aliran maklumat, replikasi dan pembaikan DNA serta pembiakan sel dibincangkan. Beberapa teknik penting dalam kajian biologi sel dan molekul juga ditekankan.

This course covers the properties of molecules that form the cells and their chemical bases, the relationships between cell structural components and functions, as well as the interactions between cells and their environment. Energy flow in cells, information flow, DNA replication and repair, as well as cell reproduction are discussed. Several important techniques in cell and molecular biology are also emphasised.

BGY3003 Biologi Perkembangan / *Developmental Biology* 3(2+1)

Prasyarat : Tiada / None

Kursus ini merangkumi konsep, prinsip dan proses perkembangan sel pembiakan dalam perkembangan embrio tumbuhan dan haiwan. Corak perkembangan dan asas genetik dalam haiwan dan tumbuhan turut dibincangkan.

This course covers the concepts, principles and development processes of reproductive cells in plant and animal embryonic development. Pattern of development and basic genetics in animals and plants are also discussed.

BGY3004 Evolusi Biologi / *Evolutionary Biology* 2(2+0)

Prasyarat : Tiada / None

Kursus ini merangkumi gambaran menyeluruh berkenaan teori evolusi moden dan menerangkan pembentukan corak biodiversiti daripada perspektif evolusi. Interaksi dengan beberapa disiplin biologi dibincangkan. Penekanan diberikan kepada kefahaman teori, konflik, tindak balas dan hubungan kait evolusi biologi dalam kehidupan.

This course covers a broad overview of the theory of modern evolution and to explain the formation of biodiversity patterns from the evolutionary perspective. Interactions with other biological disciplines are discussed. Emphasis is given to understand the theory, conflict, reactions, and relations of biological evolution in life.

BGY3005 Pengenalan kepada Bioetika / *Introduction to Bioethics* 1(1+0)

Prasyarat : Tiada / None

Kursus ini merangkumi pengetahuan biologi dan nilai-nilai manusia dalam mengkaji isu-isu etika yang muncul dengan perkembangan biologi. Kursus ini akan membincangkan teori asas dan kaedah

daripada perspektif etika, perundangan dan polisi untuk isu-isu bioetika semasa.

This course comprises the incorporation of biological knowledge and human values, and aims to study the ethical issues emerging from advances in biology. This course will address essential theories and methods from the perspective of ethics, law and policy for current bioethical issues.

BGY3100 Biologi Mikroorganisma / *Biology of Microorganisms* 3(2+1)

Prasyarat : Tiada / None

Kursus ini merangkumi diversiti, morfologi, fisiologi, pembiakan dan genetik mikroorganisma. Kepentingan dan kegunaannya dalam pelbagai bidang juga dibincangkan.

This course covers the diversity, morphology, physiology, reproduction, and genetics of microorganisms. The importance and the applications of the microorganisms in different fields are also discussed.

BGY3103 Diversiti Tumbuhan / *Plant Diversity* 4(3+1)

Prasyarat : Tiada / None

Kursus ini meliputi pengelasan moden alam Plantae dengan penekanan kepada kepelbagaian aspek morfologi, pembiakan, taburan, dan nilai komersial. Evolusi tumbuhan vaskular dibincangkan berasaskan penyesuaian kepada persekitaran daratan dan peningkatan kesempurnaan yang menjadikan mereka dominan.

This course covers a modern classification of Plantae kingdom, emphasizing the aspects of morphological, reproductive, distribution, and commercial value. The evolution of vascular plants is discussed in terms of their adaptation towards the terrestrial environment and increasing perfection, culminating in the flowering plants' dominance.

BGY3104 Diversiti Haiwan / *Animal Diversity* 4(3+1)

Prasyarat : Tiada / None

Kursus ini merangkumi kepelbagaian dan tinjauan filogeni berkaitan pengelasan taksonomi haiwan dengan penekanan diberikan kepada haiwan akuatik dan daratan yang mempunyai kepentingan ekonomi dan menjadi sumber makanan. Isu berkaitan teknik pengumpulan, pengawetan, serta pemuliharaan haiwan dibincangkan.

This course covers diversity and phylogenetic surveys related to animals' taxonomic classification with emphasizing to the local aquatic and terrestrial animals of economic importance. Issues regarding animal collection, preservation, and conservation are discussed.

BGY3201 Struktur dan Fungsi Tumbuhan / *Plant Structure and Function* 3(2+1)

Prasyarat : Tiada / None

Kursus ini meliputi kepelbagaian morfologi dan anatomi, pertumbuhan primer dan sekunder akar dan batang, serta penyesuaian organ tumbuhan. Perkembangan debunga dan pundi embrio, persenyawaan, pembentukan buah dan biji benih serta penyebarannya turut dibincangkan.

This course covers the diversity of the morphology and anatomy, primary and secondary growth of the root and stem, and adaptation of plant organs. Development of pollen and embryo sac, fertilization, formation of fruit and seed, as well as dispersal are also discussed.

BGY3202 Struktur dan Fungsi Haiwan / *Animal Structure and Function* 3(2+1)

Prasyarat : Tiada / None

Kursus ini merangkumi histologi, anatomi dan fungsi pelbagai sel, tisu, organ dan sistem haiwan. Analisis slaid histologi manusia, pembedahan ke atas spesimen vertebrat dan pengenalan struktur luaran dan dalaman organ dijalankan. Penyesuaian sistem haiwan terhadap persekitarannya turut dibincangkan.

This course comprises histology, anatomy and functions of various animal cells, tissues, organs and

systems. Analysis on human histological slides, dissection on vertebrate specimens and identification on the external and internal organ structures are carried out. Adaptation of animal systems towards their environment is also discussed.

BGY3301 Fisiologi Tumbuhan / *Plant Physiology* 4(3+1)

Prasyarat : BGY3201

Kursus ini merangkumi konsep tenaga dan hubung kait tumbuhan dengan air dan persekitaran. Proses translokasi, variasi dalam asimilasi karbon, nutrien dan pertumbuhan, serta ekspresi dan pengawalan gen dibincangkan. Masalah fisiologi yang khusus bagi tumbuhan di kawasan tropika, fisiologi stres dan penggunaan fisiologi tumbuhan dalam pertanian dan keterjaminan makanan (*food security*) ditekankan.

This course covers the concept of energy and the relationship of plants to water and the environment. Translocation processes, variations in carbon assimilation, nutrients, and growth, as well as gene expression and regulation are discussed. Physiological problems specific to plants in the tropics, stress physiology and the use of plant physiology in agriculture and food security are emphasised.

BGY3302 Fisiologi Haiwan / *Animal Physiology* 4(3+1)

Prasyarat : BGY3202

Kursus ini merangkumi konsep homeostasis sebagai asas kepada semua fenomena fisiologi dan sistem tubuh haiwan. Fokus diberikan kepada fungsi, pengawalan dan integrasi proses biologi dengan perubahan luaran dan dalaman sistem hidup. Adaptasi fisiologi terhadap cabaran persekitaran yang khusus dan unik untuk kelas haiwan tertentu juga ditekankan.

The course comprises the concept of homeostasis as a basis to all physiological phenomena and systems of the human body. Focus is given to the functions, regulations and integration of biological processes towards the internal and external changes of the living system. Physiological adaptations towards environmental challenges that are specific and unique to certain animal classes are also highlighted.

BGY3402 Prinsip Ekologi / *Principles of Ecology* 4(3+1)

Prasyarat : Tiada / *None*

Kursus ini merangkumi topik berkaitan dengan interaksi dinamik organisma dengan komponen biologi dan bukan biologi persekitaran. Isu semasa berkaitan ekologi dan alam sekitar, perubahan iklim, eksploitasi ekosistem, kesan pencemaran, pemantauan biologi dan kaedah pemuliharaan juga dibincangkan.

This course covers topics related to dynamic interactions of organisms with the biological and non-biological components of the environment. Current ecological and environmental issues, climate change, exploitation of ecosystems, pollution effects, biomonitoring and conservation methods are discussed.

BGY3501 Genetik / *Genetics* 4(3+1)

Prasyarat : BGY3002

Kursus ini merangkumi pelbagai konsep genetik. Variasi protein dan DNA, teknologi DNA rekombinan, kejuruteraan genetik, pembiakbakaan dan pengenalan bioinformatik ditekankan. Peranan genetik dalam perubatan, pertanian, pemuliharaan biodiversiti dan biosumber dibincangkan.

This course covers various genetic concepts. Protein and DNA variations, DNA recombinant technology, genetic engineering, breeding, and introduction to bioinformatics are emphasised. The roles of genetics in medicine, agriculture, biodiversity and bioresources conservation are discussed.

BGY3701 Biostatistik / *Biostatistics* 3(2+1)

Prasyarat : Tiada / *None*

Kursus ini merangkumi reka bentuk eksperimen dan kaedah analisis statistik untuk pelbagai data

biologi. Penyelesaian permasalahan kajian biologi melalui kaedah statistik turut ditekankan.

This course encompasses experimental design and methods of statistical analysis of various biological data. Solving biological research problems through statistical methods are emphasised.

BGY4902 Kaedah Penyelidikan dan Kerja Lapangan Biologi / Research 3(1+2)
Methodology and Fieldwork in Biology

Prasyarat : Tiada / None

Kursus ini merangkumi prinsip asas dan teknik saintifik dalam mereka bentuk dan melaksanakan uji kaji, mengumpul data, analisis statistik, kajian literatur dan penulisan saintifik. Teknik asas pensampelan dalam kajian biologi dan ekologi diperkenalkan. Kerja lapangan yang melibatkan beberapa ekosistem terpilih dijalankan.

This course includes basic principles and scientific techniques in designing and conducting experiment, data collection, statistical analysis, literature review and scientific writing. Basic sampling techniques used in biological and ecological studies are introduced. Fieldwork involving selected ecosystems are conducted.

BGY4903# Latihan Industri / Industrial Training 8(0+8)

Prasyarat : Dengan Kebenaran Jabatan / With the Permission of the Department

Kursus ini merangkumi pendedahan kepada persekitaran kerja yang sebenar di industri/organisasi. Aplikasi teori, praktikal, kemahiran komunikasi dan penyelesaian masalah turut ditekankan.

This course covers the exposure to real working environment in industries/organizations. Application of the theoretical, practical, communication and problem solving skills are also emphasised.

BGY4959 Disertasi Bacelor / Bachelor Dissertation 6(0+6)

Prasyarat : Dengan Kebenaran Jabatan / With the Permission of the Department

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk kajian, pengumpulan dan analisis data yang sesuai diberikan penekanan.

This course covers the preparation of proposal, implementation, and scientific writing of research project. Scientific approach to generate data systematically through appropriate research design, data collection and analysis are emphasised.

BGY4001 Evolusi dan Ekologi Perlakuan / Evolution and Behavioural Ecology 3(2+1)

Prasyarat : BGY3104

Kursus ini merangkumi konsep evolusi, ekologi dan perlakuan haiwan dan kaitannya dengan persekitaran. Teori evolusi Darwin, evolusi haiwan hidup berkumpulan dan kesannya ke atas aspek ekologi dan perlakuan haiwan dibincangkan.

This course covers the concepts of evolution, ecology, animal behaviour and their relationship with the environment. Darwin's theory of evolution, evolution of group living and the consequences on animal behaviour, and ecological aspects are emphasised.

BGY4101 Mikologi / Mycology 4(3+1)

Prasyarat : BGY3100

Kursus ini meliputi struktur sel dan perkembangan propagul pembiakan kulat, pengeluaran, penyebaran dan percambahan spora serta pengelasan kulat. Interaksi spesies dan peranan kulat dalam ekosistem, bioteknologi dan perindustrian dibincangkan.

This course covers the cell structure and development of reproductive propagules of fungi, production, dispersal and germination of spores as well as fungal classification. Species interaction and the roles of fungi in ecosystems, biotechnology and industries are discussed.

BGY4103 Biologi dan Aplikasi Vermi / *Biology and Application of Vermi* 3(2+1)

Prasyarat : BGY3104

Kursus ini merangkumi morfologi, anatomi, biologi, habitat, dan kepelbagaian spesies cacing tanah tempatan. Teknik pensampelan, pengecaman dan pengelasan cacing tanah diperkenalkan. Kepentingan cacing tanah dalam ekosistem dan aspek komersial juga dibincangkan.

This course covers morphology, anatomy, habitat and diversity of local earthworm species. Sampling techniques, identification and classification of earthworms are introduced. The importance of earthworms in ecosystem and their commercial aspects are also discussed.

BGY4105 Fikologi / *Phycology* 3(2+1)

Prasyarat : BGY3100

Kursus ini merangkumi definisi dan pengelasan alga, morfologi, perkembangan fikologi, kepentingan dan kegunaan alga, kaedah pembiakan dan kitaran hidup, fisiologi dan ekologi alga.

This course covers the definition and classification of algae, morphology, development of phycology, importance and uses of algae, method of reproduction and life cycle, physiology and algal ecology.

BGY4106 Biologi Organisma Akuatik Komersial / *Biology of Commercial Aquatic Organisms* 4(3+1)

Prasyarat : BGY3104

Kursus ini merangkumi kepelbagaian biodiversiti, morfologi, fisiologi, dan biologi organisma akuatik yang penting dari segi komersial. Perhubungan ekologi organisma akuatik dengan persekitaran ditekankan. Kepelbagaian genetik, kepentingan sosio-ekonomi, dan impak terhadap alam sekitar oleh spesies terpilih dibincangkan.

This course covers various aspects of biodiversity, morphology, physiology, and biology of commercially important aquatic organisms. Their ecological relationships with the environments are emphasised. Genetic diversity and socioeconomic importance of selected species and their impacts on environment are explained.

BGY4107 Biologi dan Propagasi Alga Komersial / *Biology and Propagation of Commercial Algae* 4(3+1)

Prasyarat : Tiada / *None*

Kursus ini merangkumi pendedahan kepada alga mikro dan makro komersial. Keperluan biologi dan fizikal untuk tujuan propagasi, produk serta potensi komersial alga dibincangkan. Kaedah propagasi, masalah yang dihadapi, teknik tuaian hasil dan teknik pemprosesan produk daripada alga diterangkan.

This course encompasses an exposure to commercial micro and macro algae. The biological and physical requirements for propagation, product and the commercial potential from the algae are discussed. Propagation methods, problems encountered, harvesting and processing techniques are explained.

BGY4108 Parasitologi dan Entomologi Kesihatan/ *Parasitology and Entomology in Health* 4(3+1)

Prasyarat : BGY3104

Kursus ini merangkumi konsep utama dalam parasitologi, morfologi, dan biologi organisma parasit termasuk vektor serangga yang berkepentingan kesihatan. Isu kesihatan manusia yang diakibatkan oleh parasit termasuk zoonosis, keimunan, epidemiologi, dan program pencegahan jangkitan parasit terkini dibincangkan.

This course covers major concepts in parasitology, morphology, and biology of parasites including insect vectors of health importance. Human health issues that are caused by these parasites, including zoonoses, immunity, epidemiology, and current parasite infection prevention programmes are discussed.

BGY4109 Biosistematik dan Pemuliharaan Tumbuhan Berbiji / *Biosystematics and Conservation of Seed Plants* 4(3+1)

Prasyarat : BGY3201

Kursus ini meliputi kepentingan dan kaedah pengelasan tumbuhan berbiji. Peranan genetik dalam biosistematik, biogeografi dan pemuliharaan dibincangkan. Status semasa, ancaman dan pemuliharaan in situ dan ex situ flora Malaysia ditekankan.

This course covers the importance and methods of classification of seed plants. The role of genetics in biosystematics, biogeography and conservation are discussed. Current status, threats, in situ and ex situ conservation of Malaysian flora are emphasised.

BGY4110 Biologi dan Biodiversiti Serangga / *Insect Biology and Biodiversity* 4(3+1)

Prasyarat : BGY3104

Kursus ini merangkumi pemahaman biologi dan kepentingannya dalam biodiversiti serangga. Order utama serangga yang berkaitan dengan pelbagai aspek entomologi pemuliharaan, ekonomi, urban, dan perubahan tropika dibincangkan.

This course covers understanding of the biology of insects and its importance in biodiversity. Major orders of insects related to conservation, economic, urban and medical entomology in the tropics are discussed.

BGY4111 Fitokimia dan Kimotaksonomi / *Phytochemistry and Chemotaxonomy* 4(3+1)

Prasyarat : BGY3201

Kursus ini merangkumi pengetahuan semasa mengenai komposisi fitokimia dalam takson tumbuhan. Taburan, kepelbagaian struktur sebatian semula jadi, kepentingan ekonomi dan peranan fitokimia tersebut dalam evolusi di pelbagai peringkat taksonomi tumbuhan dibincangkan.

This course comprises current knowledge on the phytochemical composition of plant taxa. Distribution, diversity of natural compounds structure, function, economic importance and the role of phytochemicals in the evolution of various stages of plant taxonomy are discussed.

BGY4302 Fisiologi Persekitaran Tumbuhan / *Plant Environmental Physiology* 3(2+1)

Prasyarat : BGY3301

Kursus ini merangkumi gerak balas fisiologi tumbuhan terhadap persekitaran. Pengaruh cahaya terhadap pertumbuhan dan metabolisme karbon, keperluan mineral dan air, gerak balas dan adaptasi tumbuhan terhadap stres air dan suhu, keracunan dan interaksi antara organisma dibincangkan.

This course covers the physiology of plants response to their environment. The effects of light on growth, carbon metabolism, mineral and water requirements, response and adaptation of plant to water and temperature stress, toxicity and interaction among organisms are discussed.

BGY4303 Endokrinologi Pembiakan / *Endocrinology of Reproduction* 3(2+1)

Prasyarat : Tiada / None

Kursus ini memberi penekanan kepada sistem endokrin dan pengawalaturan hormon pertumbuhan vertebrat. Peranan hormon, kompleks reseptor-hormon dan paksi hipotalamus-pituitari dalam sistem pembiakan vertebrat dibincangkan. Perkaitan faktor persekitaran dan endocrine disruptors dalam pengawalaturan hormon endokrin juga ditekankan. Pelajar juga didedahkan kepada kemahiran pensampelan mengikut etika haiwan yang ditetapkan.

This course emphasizes on endocrine system and regulation of growth hormones in vertebrates. Role of hormones, receptor-hormone complexes and hypothalamic-pituitary axis in vertebrate reproductive system are discussed. Interaction between environmental factors and endocrine disruptors in endocrine hormone regulation are also emphasised. Students are also exposed to sampling skills according to established animal ethics.

BGY4304 Neurotoksikologi Perkembangan / *Developmental Neurotoxicology* 3(2+1)

Prasyarat : Tiada / None

Neurotoksikologi perkembangan merangkumi pelbagai disiplin termasuk fisiologi, genetik, anatomi, toksikologi dan ekologi. Fokus utama adalah untuk memahami kesan keneurotoksikan semasa perkembangan sistem saraf embrio dan fetus. Penyakit yang berkaitan dengan gangguan perkembangan sistem saraf dan peranan faktor persekitaran dalam etiologi penyakit dibincangkan. Garis panduan bahan kimia bersesuaian dengan teknik penilaian risiko sedia ada ditekankan.

Developmental neurotoxicology encompasses various disciplines such as physiology, genetics, anatomy, toxicology and ecology. The focus is to understand the impact of neurotoxicity on the development of embryonic and fetal nervous system. Diseases associated with nervous system development and the role of environmental factors in the etiology of these diseases will be discussed. Guidelines for chemicals in accordance with the existing risk assessment techniques are emphasised.

BGY4305 Prinsip dan Kaedah Epidemiologi / *Principles and Methods of Epidemiology* 3(2+1)

Prasyarat : Tiada / None

Kursus ini merangkumi konsep asas, prinsip dan kaedah epidemiologi. Pendekatan umum kursus ini adalah teori dan kuantitatif dengan memberi tumpuan kepada kaedah menjalankan kajian tentang etiologi penyakit. Penekanan diberikan kepada reka bentuk kajian epidemiologi asas, pensampelan, penentuan saiz sampel, bias dalam pemilihan, teknik pengumpulan data, sumber data sekunder dan pengenalan kepada model matematik dalam epidemiologi.

This course covers basic concepts, principles and methods of epidemiology. The general approach of this course is theoretical and quantitative, focusing on the methods in conducting research investigating the aetiology of the disease. Emphasis is placed on basic epidemiological study designs, sampling, sample size determination, the bias in selection, data collection techniques, secondary data sources and an introduction to mathematical models in epidemiology.

BGY4306 Komunikasi Kimia Serangga / *Insect Chemical Communication* 4(3+1)

Prasyarat : Tiada / None

Kursus ini merangkumi peranan komunikasi kimia dalam ekologi kimia serangga. Pemahaman tentang manipulasi komunikasi kimia serangga diberikan dalam konteks kelakuan serangga. Aplikasi praktikal semiokimia dalam pengurusan serangga perosak dibincangkan.

This course covers the role of chemical communication in insect chemical ecology. An understanding of manipulating insect chemical communication is provided in the context of insect behaviour. Practical applications of semiochemicals in managing pest insects are discussed.

BGY4401 Ekologi Hutan Tropika / *Tropical Forest Ecology* 4(3+1)

Prasyarat : BGY3402

Kursus ini merangkumi persekitaran, iklim dan mikroiklim, struktur, fungsi, biodiversiti dan komponen pelbagai jenis hutan tropika. Kesan aktiviti manusia terhadap pemuliharaan dan kemapanan hutan juga ditekankan.

This course covers environment, climate and microclimate, structure, functions, biodiversity and components of various types of tropical forests. Impacts of human activities on forest conservation and sustainability are also emphasised.

BGY4402 Ekologi Hidupan Liar / *Wildlife Ecology* 4(3+1)

Prasyarat : BGY3402

Kursus ini merangkumi konsep dan ekologi hidupan liar. Penentuan kepadatan hidupan liar, ancaman dan aspek ekologi dalam pemuliharaan haiwan liar di kepulauan hutan, taman negara dan taman laut di Malaysia dibincangkan.

This course covers the concepts and ecology of wildlife. Wildlife density estimation, threats and ecological aspects in the conservation of wildlife in forest islands, national and marine parks in Malaysia are discussed.

BGY4403 Ekotoksikologi / *Ecotoxicology* 4(3+1)

Prasyarat : BGY3402

Kursus ini merangkumi sumber dan kimia bahan toksik. Pengekstrakan, pengesanan, biocerakan dan tindak balas organisma terhadap bahan toksik dibincangkan. Kesan bahan toksik terhadap organisma, populasi dan komuniti di pelbagai ekosistem ditekankan.

This course covers the source and chemistry of toxicants. Extraction, detection, bioassay and response of organisms towards toxicants are discussed. Effects of toxicants on the organisms, populations and communities in various ecosystems are emphasised.

BGY4404 Limnologi dan Oseanografi / *Limnology and Oceanography* 4(3+1)

Prasyarat : BGY3402

Kursus ini merangkumi penggunaan air dalam pengangkutan, adaptasi struktur organisma, produktiviti primer dan sekunder dan organisma pengurai. Perubahan bermusim, enapan dan pengendapan, stratifikasi mikro, kesan aktiviti manusia ke atas sistem akuatik dan perkembangan semasa dalam bidang limnologi dan biologi marin diberikan penekanan.

This course encompasses the use of waters in transportation, adaptive features of organisms, primary and secondary productivity, and decomposers. Seasonal variations, sediment and sedimentation, microstratification, impact of man on aquatic system, and recent development in the field of limnology and marine biology are highlighted.

BGY4405 Bakteriologi Persekitaran / *Environmental Bacteriology* 3(2+1)

Prasyarat : BGY3100

Kursus ini merangkumi konsep, pengetahuan dan aplikasi berkaitan bakteria yang berhubung kait dengan kehidupan manusia dan persekitaran. Teknik pemencilan, pencegahan jangkitan, pengurusan toksin dan aplikasinya dalam persekitaran juga diperkenalkan. Hubung kait antara aktiviti bakteria dengan kegunaan dan ancaman kepada persekitaran juga dibincangkan.

This course covers concepts, knowledge and application on bacteria related to mankind and environment. Isolation technique, infection prevention, toxin management and their applications in environment are introduced. Relationship between bacterial activities towards the usage and threat on environment are also discussed.

BGY4406 Biologi dan Ekologi Rumput Laut / *Biology and Ecology of Seagrasses* 4(3+1)

Prasyarat : Tiada / *None*

Kursus ini merangkumi morfologi, anatomi, taksonomi dan kepelbagaian spesies rumput laut. Teknik pensampelan, pengecaman, pengawetan dan penyediaan herbarium dijalankan. Taburan dan biogeografi rumput laut berkaitan dengan pelbagai habitat, dan ciri penyesuaian hidup kepada persekitaran marin, pertumbuhan, produktiviti dan faktor utama yang mengawalnya dibincangkan.

This course covers the morphology, anatomy, taxonomy and species diversity of seagrasses. Sampling techniques, identification, preservation and herbarium preparation are carried out. Distribution and biogeography of seagrasses in relation to various habitats and adaptive characteristics to marine environment, growth, productivity and primary factors controlling them are discussed.

BGY4408 Limnologi Gunaan / *Applied Limnology* 4(3+1)

Prasyarat : BGY3402

Kursus ini merangkumi aras trofik sistem akuatik dan faktor yang mempengaruhi perubahannya. Penggunaan ekoteknologi dan biomanipulasi dalam pemulihan ekosistem tercemar serta pengurusan sistem akuatik air tawar ditekankan.

This course covers the trophic levels of aquatic systems and factors controlling their changes. The use of ecotechnology and biomanipulation techniques in recovering polluted ecosystems as well as management of inland aquatic systems are emphasised.

BGY4409 Pengurusan dan Pemuliharaan Ekosistem Akuatik / *Aquatic Ecosystem Management and Conservation* 4(3+1)

Prasyarat : BGY3402

Kursus ini merangkumi organisasi struktur ekosistem tasik, sungai, empangan, muara dan laut. Pengaliran tenaga dan kitaran nutrien melalui populasi dan komuniti serta strategi pembangunan dan pengurusan mapan ekosistem akuatik juga dibincangkan.

This course covers the structural organization of lake, river, reservoir, estuary and marine ecosystems. Energy flow and nutrient cycling through populations and communities as well as strategies for sustainable development and management of aquatic ecosystem are also discussed.

BGY4501 Polimorfisme Genetik / *Genetic Polymorphisms* 4(3+1)

Prasyarat : BGY3501

Kursus ini merangkumi asas genetik dalam variasi di tahap polimorfisme dan kaitannya dengan kehidupan serta kepentingannya dalam evolusi. Teknik mengesan pelbagai bentuk dan kegunaan polimorfisme dalam perubatan, pertanian, biosistematik, pemuliharaan, evolusi, antropologi dan forensik dibincangkan.

This course covers the genetic basis of variation found at polymorphic levels and their relevance to life as well as their significance in evolution. Techniques to identify the various forms and the use of polymorphisms in medicine, agriculture, biosystematics, conservation, evolution, anthropology, and forensics are discussed.

BGY4502 Pembiakbakaan dan Genetik / *Breeding and Genetics* 4(3+1)

Prasyarat : BGY3501

Kursus ini merangkumi prinsip genetik termasuk sitogenetik, genetik kualitatif dan kuantitatif serta prinsip pembiakbakaan semulajadi dan aruhan. Program pemilihan dan teknik manipulasi kromosom dalam organisma terpilih dibincangkan.

This course covers the principles of genetics including cytogenetics, qualitative and quantitative genetics and principles of natural and induced breeding. Selection programme and chromosome manipulation techniques in selected organisms are discussed.

BGY4503 Biologi Pembiakan Bandingan / *Comparative Reproductive Biology* 4(3+1)

Prasyarat : BGY3104

Kursus ini merangkumi prinsip biologi pembiakan bandingan bagi kumpulan vertebrat utama. Penggunaan kaedah cerapan kuantitatif, histologi dan mikroskop dalam pemerhatian dan pengenalpastian peringkat perkembangan gonad dan embrio dijalankan.

This course covers principles of comparative reproductive biology of major groups of vertebrates. The use of various quantitative and histological methods and microscopy to describe, observe and identify the developmental stages of gonad and embryo are carried out.

BGY4504 Genetik Populasi / *Population Genetics* 4(3+1)

Prasyarat : BGY3501

Kursus ini merangkumi keseimbangan Hardy-Weinberg, evolusi dan analisis genetik populasi. Faktor evolusi yang boleh mengubah struktur genetik populasi dan kesannya ke atas populasi dibincangkan. Genetik populasi molekul dan filogenetik juga dibincangkan.

This course covers the Hardy-Weinberg equilibrium, evolution and population genetic analysis. Evolutionary forces that can change the genetic structure of population and the effects on the population are discussed. Molecular population genetics and phylogenetics are also discussed.

BGY4505 Genetik Kuantitatif / *Quantitative Genetics* 4(3+1)

Prasyarat : BGY3501

Kursus ini merangkumi konsep genetik kuantitatif. Analisis statistik dan analisis biometrik berkaitan dengan genetik kuantitatif juga dibincangkan.

This course covers concepts of quantitative genetics. Statistical analysis and biometrical analysis relating to quantitative genetics are also discussed.

BGY4506 Bioinformatik dalam Analisis Gen dan Protein / *Bioinformatics in Analysis of Genes and Proteins* 3(2+1)

Prasyarat : BGY3501

Kursus ini merangkumi konsep, kaedah, dan perisian yang digunakan dalam bioinformatik. Pembelajaran perisian *perl*, pemeriksaan kualiti penjujukan dan penetapan saluran kerja aliran penyelidikan juga ditekankan.

This course covers the concepts, methods, and tools used in bioinformatics. Learning perl, checking sequencing quality and establishing a research workflow pipeline are also emphasised.

BGY4507 Genetik Pemuliharaan Biosumber / *Conservation Genetics of Bioresources* 4(3+1)

Prasyarat : BGY3501

Kursus ini memberikan latar belakang mengenai prinsip genetik pemuliharaan untuk mengenal pasti dan menganggar ancaman genetik terhadap biosumber. Kegunaan kaedah genetik dalam pemuliharaan dan pengurusan biosumber dibincangkan.

This course provides a background on the principles of conservation genetics to identify and quantify genetic threats to bioresources. The usefulness of genetic tools in conservation and management of bioresources are discussed.

BGY4508 Kemajuan Molekular dalam Biologi Tumbuhan / *Molecular Advances in Plant Biology* 3(3+0)

Prasyarat : BGY3501

Kursus ini memberi tumpuan kepada teori dan kemajuan dalam pelbagai bidang bioteknologi tumbuhan. Maklumat yang dikumpul melalui aplikasi teknologi ini dapat memberikan pemahaman yang lebih jelas mengenai jaringan kompleks yang mengawal atur laluan utama dalam sistem tumbuhan.

This course focuses on the theory and advances in various fields of plant biotechnology. The information gathered through the applications of these technologies provides deeper insights into the complex networks that regulate important pathways in plants.

BGY4801 Teknik Pengasingan dan Penulenan dalam Analisis Protein / *Separation and Purification Techniques in Protein Analysis* 3(2+1)

Prasyarat : Tiada / None

Kursus ini merangkumi konsep analisis kromatografi dan elektroforesis. Teknik pengasingan dan penulenan menggunakan kromatografi penukaran ion, kromatografi pemfokusan, penurasan gel, analisis elektroforesis dan pemfokusan isoelektrik dibincangkan.

The course covers the analytical concept of chromatography and electrophoresis. Separation and purification techniques using ionic exchange chromatography, chromatofocusing, gel filtration, electrophoresis and isoelectric focusing techniques are discussed.

Nota / Note:

Ditawarkan untuk pelajar BSBDK sahaja / *Offered for BSBDK students only*

*Ditawarkan untuk pelajar BSBDP(K) sahaja / *Offered for BSBDP(K) students only*

SINOPSIS KURSUS/ COURSE SYNOPSIS

Jabatan Fizik/ Department of Physics

PHY3000 Prinsip Fizik/ *Principles of Physics* 4(3+1)

Prasyarat : Tiada

Kursus ini memperkenalkan prinsip asas fizik yang merangkumi mekanik dan dinamik, bendalir, haba, cahaya, gelombang, elektrik, magnet, elektronik dan keradioaktifan. Perbincangan dijalankan secara kualitatif dan kuantitatif. Penggunaan prinsip asas fizik dalam kehidupan seharian turut diberi penekanan.

This course introduces the basic principles of physics covering mechanics and dynamics, fluid, heat, light, waves, electricity, magnetism, electronic and radioactivity. Discussion is done qualitatively and quantitatively. Application of the basic principles of physics to society is also emphasized.

PHY3001 Mekanik/ *Mechanics* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi pergerakan zarah dan jasad tegar di dalam satu, dua dan tiga dimensi berdasarkan Hukum Newton. Topik tentang bendalir, getaran dan gelombang mekanik serta aplikasinya turut dibincangkan.

This course covers the motion of particles, and rigid bodies in one, two and three dimensions based on Newton's Law. Topics on fluid, vibration and wave mechanics as well as their application are also discussed.

PHY3002 Keelektrikan dan Kemagnetan/ *Electricity and Magnetism* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi konsep umum dalam menyelesaikan masalah fizik melibatkan keelektrikan dan kemagnetan. Hukum-hukum penting dalam teori elektrik dan magnet turut ditekankan.

This course covers general concepts of solving physics problems that involved electricity and magnetism. Important laws in electric and magnetic theory are also emphasized.

PHY3103 Fizik I/ *Physics I* 4(3+1)

Prasyarat : Tiada

Kursus ini merangkumi pergerakan zarah dan jasad tegar di dalam satu, dua dan tiga dimensi berdasarkan Hukum Newton. Topik tentang getaran, gelombang mekanik serta hukum termodinamik dan aplikasinya dalam fizik terma turut dibincangkan.

This course covers the motion of particles, and rigid bodies in one, two and three dimensions based on Newton's Law. Topics on vibration, wave mechanics and laws of thermodynamics and their application in thermal physics are also discussed.

PHY3200 Fizik Bahan/ *Materials Physics* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi pengetahuan asas mengenai pengelasan bahan dan penciriannya. Struktur hablur, kecacatan, kemeresapan, pemejatan dan sifat bahan mekanikal, elektrik, terma, optik dan magnet juga dibincangkan.

This course covers fundamental knowledge on the classification of materials and their properties. The crystal structure, defect, diffusion, solidification, and materials properties of mechanical, electrical,

thermal, optical and magnetics are discussed.

PHY3201 Fizik Keadaan Pepejal/ *Solid State Physics* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkum struktur dan daya ikatan hablur. Kesan getaran kekisi dan model elektron bebas ke atas sifat terma, mod akustik dan optik dalam logam dibincangkan. Model jalur tenaga digunakan untuk membezakan semikonduktor, penebat dan logam. Jenis kecacatan dalam logam juga dibincangkan.

This course covers crystal structure and crystal binding forces. Effect of lattice vibration and free electron model on thermal, acoustic and optical modes in metals are discussed. Energy band model is employed to distinguish semiconductor, insulator and metals. Types of defects in metals are also discussed.

PHY3208 Kemagnetan dan Bahan Magnet/ *Magnetism and Magnetic Materials* 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkumi konsep asas kemagnetan dan pengelasan bahan magnet. Teori umum berasaskan fizik klasik, mekanik kuantum dan klasifikasi jenis bahan magnet ditekankan. Perkembangan dan aplikasi bahan magnet juga dibincangkan.

This course covers the basic concept of magnetism and types of magnetic materials. General theories based on classical physics, quantum mechanics, and classification of magnetic materials are emphasized. The development and application of magnetic materials are also discussed.

PHY3209 Termodinamik/ *Thermodynamics* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi konsep fizik terma dan hukum termodinamik. Asas mekanisme pemindahan haba seperti pengaliran haba, perolakan dan radiasi juga dibincangkan.

The course covers the concept of thermal physics and thermodynamics law. The basics of heat transfer mechanisms such as heat conduction, convection and radiation are also discussed.)

PHY3303 Sensor dan Transduser/ *Sensors and Transducers* 4(3+1)

Prasyarat : PHY3306

Kursus ini merangkumi prinsip dan sifat fizikal bagi pelbagai jenis sensor dan transduser. Parameter sensor bagi pengukuran kuantiti asas fizik dan pengenalan kepada pelbagai jenis sensor terkini dibincangkan.

This course covers the principles and physical properties of sensors and transducers. Sensor parameters for measurement of fundamental quantities in physics and recent developments in sensor fields are discussed.

PHY3304 Prinsip Sistem Pengukuran/ *Principles of Measurement Systems* 4(3+1)

Prasyarat : Tiada

Kursus ini merangkumi ciri dan unsur asas sistem pengukuran. Prinsip umum berkaitan ciri statik dan dinamik, kesan hingar dan piawaian dalam industri dan makmal penyelidikan dibincangkan.

This course covers characteristic and basic elements in the measurement system. General aspects of measurement systems such as static and dynamic characteristics, effects of noise and measurement systems standard in the industry and research laboratory are discussed.

PHY3306 Elektronik/ *Electronics* 4(3+1)

Prasyarat : Tiada

Kursus ini merangkumi analisis litar elektronik arus terus dan ulang alik. Prinsip asas, konsep teknikal dan kegunaan peranti elektronik analog dan digital dibincangkan.

This course covers the analysis of direct and alternating current electronic circuits. The characteristics, technical concept and applications of analogue and digital electronic devices are discussed.

PHY3400 Optik/ Optics 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi prinsip dan teori cahaya sebagai satu sinar geometri dan sifat gelombang cahaya. Fenomena alam dan perkembangan teknologi berkaitan dengan optik turut dibincangkan.

This course covers the principles and theory of light as a geometrical ray and wave nature of light. The nature phenomenon and technology development of optics are also discussed.

PHY3401 Keelektromagnetan/ Electromagnetism 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi hukum asas, saling tindakan dan teknik penyelesaian masalah melibatkan ruang elektrostatik dan magnetostatik. Medan elektromagnet berubah terhadap masa yang dikaitkan dengan persamaan Maxwell turut ditekankan.

This course covers basic laws, interactions and problem-solving techniques involving electrostatics and magnetostatics spaces. Time-varying electromagnetics field which is related to Maxwell's equations are also discussed.

PHY3600 Kaedah Matematik dalam Fizik I/ Mathematical Methods in Physics I 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi teknik matematik asas dan formulasi penyelesaian bermatematik melibatkan pelbagai sistem fizik. Konsep matriks, aljabar kompleks, siri Fourier dan persamaan pembezaan turut dibincangkan.

This course covers basic mathematical techniques and mathematical solving formulation which involves the various physics system. The concept of matrices, complex algebra, Fourier series and differential equation are also discussed.

PHY3601 Mekanik Kuantum/ Quantum Mechanics 3(3+0)

Prasyarat : PHY3105

Kursus ini merangkumi teori dan formulasi mekanik kuantum, dan persamaan Schrodinger dalam sistem satu, dua dan tiga dimensi. Teknik formulasi operator dalam teori momentum sudut juga turut ditekankan.

This course covers introduction and formulation of quantum mechanics and Schrödinger equation in the systems of one, two and three dimensions. Operator formulation technique in orbital momentum theory is also emphasized.

PHY3602 Mekanik Statistik/ Statistical Mechanics 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi sifat cirian sistem makroskopik dan mikroskopik sistem fizik. Konsep kebarangkalian, ensembel dan kaitannya dengan sifat statistik sistem zarah juga ditekankan.

This course covers the characteristic features of macroscopic and microscopic in physics system. The concept of probability, the ensemble and their relation to particles statistical behaviour of the system are also emphasized.

PHY3603 Mekanik Klasik/ Classical Mechanics 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi pergerakan zarah dalam satu, dua dan tiga dimensi. Konsep kelengkungan dan formalisme mekanik Lagrangian dan Hamiltonian turut dibincangkan.

This course covers the motion of a particle in one, two and three dimensions. Trajectory concept and Lagrangian and Hamiltonian mechanics formalism are also discussed.

PHY3701 Astronomi/ Astronomy 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi konsep fenomena astronomi, pembentukan, saling tindakan dan evolusi sistem suria. Peralatan cerapan, bintang, galaksi dan kosmologi turut ditekankan.

This course covers the concepts of astronomical phenomenon, formation, interaction and evolution of solar system. The observation instruments, stars, galaxies and cosmology are also emphasized.

PHY3901 Amali Mekanik dan Fizik Terma/ *Mechanics and Thermal Physics Laboratory* 1(0+1)

Prasyarat : Tiada

Kursus ini merangkumi eksperimen mekanik, gelombang dan haba. Pengendalian peralatan makmal dalam teknik sains eksperimen yang berkaitan turut ditekankan.

This course covers the mechanics, wave and heat experiments. The handling of laboratory equipment in related experimental science techniques are also emphasized.

PHY3902 Amali Keelektromagnetan dan Optik/ *Electromagnetism and Optics Laboratory* 1(0+1)

Prasyarat : Tiada

Kursus ini merangkumi eksperimen elektrik, magnet dan optik. Pengendalian peralatan makmal dalam teknik sains eksperimen yang berkaitan turut ditekankan.

This course covers the electricity, magnetism and optics experiments. The handling of laboratory equipment in related experimental science techniques are also emphasized.

PHY3903 Pendedahan Awal Kepada Industri/ *Early Exposure to Industry* 1(0+1)

Prasyarat : Dengan kebenaran jabatan

Kursus ini merangkumi pendedahan awal tentang struktur, cara kerja dan keselamatan di tempat kerja. Pendedahan kepada proses permohonan dan etika di industri turut dibincangkan.

This course covers early exposure on structure, work procedure and safety at workplace. Exposure to the application processes and ethics at the industry are also discussed.

PHY3904 Pembelajaran Perkhidmatan Dalam Fizik/ *Service Learning in Physics* 1(0+1)

Prasyarat : PHY3001 dan PHY3002

Kursus ini merangkumi aktiviti bagi meningkatkan kesedaran dan kefahaman komuniti atau industri tentang kepentingan fizik dalam kehidupan harian. Pengetahuan fizik diaplikasikan melalui aktiviti kerjasama dengan masyarakat luar.

This course encompasses activities to increase awareness and understanding of the community or industry on the importance of physics in daily life. Knowledge in physics is applied through joined activities with the community.

PHY3921 Sintesis dan Perincian Bahan I/ *Materials Synthesis and Characterization I* 1(0+1)

Prasyarat : PHY3200

Kursus ini merangkumi eksperimen sintesis dan teknik pemprosesan bahan. Pengendalian peralatan makmal dalam sintesis dan pencirian sifat struktur, terma dan mekanik turut ditekankan.

This course covers synthesis experiments and materials processing techniques. Handling of laboratory equipment in the synthesis and characterization of structural, thermal and mechanical properties are also emphasized.

PHY3922 Sintesis dan Perincian Bahan II/ *Materials Synthesis and Characterization II* 1(0+1)

Prasyarat : PHY3200

Kursus ini merangkumi eksperimen sintesis dan teknik pemprosesan bahan. Pengendalian peralatan makmal dalam sintesis dan pencirian sifat magnetik, elektrik dan optik turut ditekankan.

This course covers synthesis experiments and materials processing techniques. Handling of laboratory equipment in the synthesis and characterization of magnetic, electrical and optical properties are also emphasized.

PHY3923 Fabrikasi dan Perincian Filem/ *Fabrication and Characterization of Films* 1(0+1)

Prasyarat : PHY3200

Kursus ini merangkumi eksperimen fabrikasi filem. Pengendalian peralatan makmal dalam fabrikasi dan pencirian filem turut ditekankan.

This course covers films fabrication experiments. Handling of laboratory equipment in the fabrication and characterization of films are also emphasized.

PHY4201 Fizik Keadaan Pepejal Lanjutan/ *Advanced Solid State Physics* 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkumi fenomena makroskopik bahan pepejal berdasarkan sifat skala atom. Penghampiran, model dan kaedah piawai untuk menjelaskan kelakuan pepejal turut dibincangkan.

This course covers macroscopic phenomena of solid materials based on the atomic scale properties. Standard approximations, models and methods for explaining the behaviour of solids are discussed.

PHY4202 Peranti Semikonduktor/ *Semiconductor Devices* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi operasi peranti semikonduktor berdasarkan struktur tenaga jalur dan statistik pembawa. Konsep persimpangan P-N dan simpang logam-semikonduktor dalam peranti semikonduktor turut dibincangkan.

This course covers the operation of semiconductor devices based on energy bands structure and carrier statistics. Concept of P-N junctions and metal-semiconductor junctions in semiconductor devices are also discussed

PHY4203 Sains Bahan/ *Materials Science* 3(3+0)

Prasyarat : PHY3103 dan PHY3104

Kursus ini merangkum konsep asas berkenaan sintesis/fabrikasi, pencirian dan potensi aplikasi bahan termaju. Ini termasuk bahan nano, komposit, bio-bahan, seramik, fotonik, polimer, bahan untuk kelestarian tenaga dan bahan pintar.

This course covers the basic concepts on synthesis/fabrication, characterisation and potential applications of advanced materials. These include nano materials, composites, bio-materials, ceramics, photonics, polymers, materials for energy sustainability and smart materials.

PHY4207 Teknologi Pemprosesan Bahan/ *Materials Processing Technology* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi kaedah umum pemprosesan bahan berasaskan logam, seramik, polimer dan

komposit. Potensi keusahawan yang berkaitan turut dibincangkan.

This course covers the common processing methods of metals, ceramics, polymers and composites. The related entrepreneurial potentials also discussed.

PHY4209 Bahan Termaju/ *Advanced Materials* 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkum konsep asas berkenaan sintesis/fabrikasi, pencirian dan potensi aplikasi bahan termaju. Ini termasuk bahan nano, komposit, bio-bahan, seramik, fotonik, polimer, bahan untuk kelestarian tenaga dan bahan pintar.

This course covers the basic concepts on synthesis/fabrication, characterisation and potential applications of advanced materials. These include nano materials, composites, bio-materials, ceramics, photonics, polymers, materials for energy sustainability and smart materials.

PHY4210 Teknologi Semikonduktor/ *Semiconductor Technology* 3(3+0)

Prasyarat : PHY4202

Kursus ini merangkumi proses fabrikasi teknologi dan pencirian bahan semikonduktor. Perkembangan proses fabrikasi terkini semikonduktor turut dibincangkan.

This course covers fabrication technology process and characterisation of semiconductor material. The development of current semiconductor fabrication process is also discussed.)

PHY4211 Analisis Struktur dan Mikrostruktur/ *Structure and Microstructure Analysis* 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkumi kaedah menganalisis struktur dan mikrostruktur bahan pejal. Pembelauan sinar-X, mikroskopi penduga dan mikroskopi elektron turut dibincangkan.

This course covers the analytical methods of structure and microstructure of solid compound. X-ray diffraction, probe microscopy and electron microscopy are also discussed.

PHY4212 Keramik, Polimer dan Komposit/ *Ceramic, Polymer and Composites* 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkumi pengkelasan jenis bahan, sifat mekanik bahan, gambarajah fasa, penghabluran dan mikrostruktur. Penggunaan bahan seramik, polimer dan komposit turut dibincangkan.

This course covers the classification of materials, mechanical properties, phase diagrams, crystallization and microstructures. The application of ceramics, polymers and composites are also discussed.

PHY4213 Metalurgi/ *Metallurgy* 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkumi struktur dan sifat bahan, fokus utama pada logam dan aloi. Konsep reka bentuk aloi dan kejuruteraan mikrostruktur turut dibincangkan.

This course covers the structure and properties of materials, focusing primarily on metals and alloys. The concepts of alloy design and microstructural engineering are also discussed.

PHY4214 Sains Kakisan/ *Corrosion Science* 3(3+0)

Prasyarat : Tiada

Kursus ini menerangkan fenomena kakisan bahan, bentuk kakisan, pencegahan dan pengurusan

kakisan. Potensi keusahawan berkaitan kakisan turut dibincangkan.

This course describes material corrosion phenomena, forms of corrosion, prevention and management of corrosion. The potential of corrosion-related entrepreneurs are discussed.

PHY4215 Bahan Nano/ Nanomaterials 3(3+0)

Prasyarat : PHY3200

Kursus ini merangkumi prinsip asas nanosains dan nanoteknologi termasuk sintesis dan fabrikasi, sifat serta penciriannya. Aplikasi bahan nano dalam teknologi terkini turut dibincangkan.

This course covers the basic principles of nanoscience and nanotechnology including the synthesis and fabrication, properties and characterization. Applications of nanomaterials in recent technology are also discussed.

PHY4216 Sains dan Teknologi Bahan Termaju/ Science and Technology of Advanced Materials 3(3+0)

Prasyarat : PHY3200

Kursus ini merangkumi pemahaman bahan termaju terkini dan pembangunan teknologi kecerdasan buatan (AI). Potensi keusahawan dalam bidang berkaitan turut dibincangkan.

This course covers the understanding of latest advanced materials and development of artificial intelligence (AI) technologies. The entrepreneurial potential in related fields are also discussed.

PHY4217 Sains dan Teknologi Superkonduktor/ Science and Technology of Superconductor 3(3+0)

Prasyarat : PHY3201

Kursus ini merangkumi bahan superkonduktor, teknologi dan aplikasi yang berkaitan. Teori kuantum mikroskopik dan fenomenologi dibincangkan.

This course covers superconducting materials, technology and related applications. Microscopic quantum theory and phenomenology are discussed.

PHY4301 Mikroprosesor & Mikrokomputer/ Microprocessors and Microcomputers 3(3+0)

Prasyarat : PHY3306

Kursus ini merangkumi seni bina mikrokomputer dan mikroprosesor. Perkakasan mikrokomputer, teknik perisian emulator, teknik pengantaramukaan dan perbandingan mikroprosesor dibincangkan.

This course covers microcomputer and microprocessor architecture. Microcomputer hardware, software emulator techniques, interfacing techniques and comparison of microprocessors are discussed.

PHY4302 Rekabentuk Peralatan Elektronik/ Design of Electronic Equipment 4(3+1)

Prasyarat : PHY3306

Kursus ini merangkumi pelbagai teori reka bentuk elektronik yang merangkumi perancangan reka bentuk sehingga proses dokumentasi. Langkah untuk mereka bentuk projek dan menghasilkan prototaip ringkas peralatan elektronik dibincangkan.

This course covers various theory of electronic designs which include design planning up to the final documentation. Steps to design a project and to produce simple prototype of electronic equipment are discussed.

PHY4303 Pengantaramukaan Komputer dan Kawalan/ Computer Interfacing and 4(3+1)

Tarikh kemaskini : 29/08/2025

Control

Prasyarat : PHY3306

Kursus ini merangkumi konsep asas dan teknik perolehan data (DAQ) untuk pensampelan isyarat elektrik serta fenomena fizikal. Kaedah pemproses, produktiviti, paparan dan kebolehan penghantaran komputer industri piawai dalam penyelesaian pengukuran efektif dibincangkan.

This course covers the basics of data acquisition (DAQ) concepts and techniques for sampling electrical as well as physical phenomenon signals. Methods of processing, productivity, display and connectivity capabilities of industry standard computer for effective measurement solution are discussed.

PHY4304 Sistem Mikropengawal dan Rekabentuk/ *Microcontroller System and Design* 4(3+1)

Prasyarat : PHY3303

Kursus ini merangkumi asas perkakasan, perisian dan integrasi sistem berasaskan mikropengawal. Seni bina mikropengawal, teknik pengaturcaraan, sistem pembangunan, reka bentuk aplikasi dan sistem mikropengawal dibincangkan.

This course covers the fundamentals of the hardware, software and integration of a microcontroller based system. Microcontroller architecture, programming techniques, development systems, application design and microcontroller systems are discussed.

PHY4305 Instrumentasi Lanjutan/ *Advanced Instrumentation* 3(3+0)

Prasyarat : PHY3304

Kursus ini merangkumi fizik asas yang berkaitan dengan pembangunan instrumentasi moden serta pelbagai aspek reka bentuk eksperimen, pengendalian maklumat dan analisis data. Peralatan khas berkaitan dengan teknik ujian tanpa musnah dan analisis terma dibincangkan.

This course covers the fundamental physics that underlies many modern instrumentation and several aspects of experimental design, information handling and data analysis. Special instruments related to non-destructive techniques and thermal analysis are discussed.

PHY4306 Elektronik Lanjutan/ *Advanced Electronics* 3(3+0)

Prasyarat : PHY3306

Kursus ini merangkumi konsep teori komponen dan peranti elektronik analog dan digital. Fungsi dan aplikasi komponen dan peranti dalam instrumentasi turut dibincangkan.

This course covers the theoretical aspect of electronic components and devices in both analogue and digital form. Function and application of the component and device in instrumentation are also discussed.

PHY4307 Sistem Kawalan Instrumentasi/ *Instrumentation Control Systems* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi kefahaman asas sistem kawalan automasi moden. Teknik, strategi kawalan, jenis pengawal dan keselamatan dibincangkan.

This course covers the fundamental understanding of modern automatic control systems. Techniques, control strategies, controller types and safety are discussed.

PHY4308 Prinsip dan Instrumentasi Ujian Tanpa Musnah/ *Principles and Instrumentation of Non-Destructive Testing* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi teori dan kaedah ujian tanpa musnah seperti elektromagnet, visual dan optik. Aplikasi teknik berkenaan di industri dibincangkan.

This course covers theory and technique of non-destructive testing such as electromagnet, visual and optic. Application of the techniques in related industries are discussed.

PHY4309 Instrumentasi Pengimejan Perubatan/ *Medical Imaging Instrumentation* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkumi teknik pengimejan perubatan, instrumentasi dalam pembentukan dan pemprosesan imej. Pemahaman mengenai konsep fizik dalam teknik pengimejan dan aplikasi klinikal dibincangkan.

This course covers the techniques of medical imaging as well as the instrumentation in image formation and processing. Understanding of the physics concept in imaging techniques and their clinical applications are discussed.

PHY4401 Keelektromagnetan Gunaan/ *Applied Electromagnetism* 3(3+0)

Prasyarat : PHY3401

Kursus ini merangkumi aplikasi persamaan Maxwell dan perambatan gelombang elektromagnet serta penggunaannya dalam teknologi terkini. Talian penghantaran, pandu gelombang, prinsip elektromagnet dalam fotonik dan antena juga turut dibincangkan.

This course covers the application of Maxwell's equations and propagation of plane waves together with current technologies. Transmission lines, waveguides, electromagnetic principles in photonics and antennas are discussed.

PHY4404 Optoelektronik dan Fotonik/ *Optoelectronics and Photonics* 3(3+0)

Prasyarat : PHY4403

Kursus ini merangkumi konsep asas optoelektronik dan fotonik. Ia merangkumi topik am di dalam optoelektronik dan fotonik termasuk pandu-gelombang dielektrik, sumber dan pancaran cahaya, peranti cahaya dan pengesanan cahaya.

This course covers fundamental concepts in optoelectronics and photonics. It covers general topics in optoelectronics and photonics including wave dielectric waveguides, light source and emission, light devices and light detectors.

PHY4405 Optik Lanjutan dan Fizik Laser/ *Advanced Optics and Laser Physics* 3(3+0)

Prasyarat : PHY3400

Kursus ini merangkumi pengetahuan dan aspek lanjutan bidang optik yang menjurus kepada aplikasi dan penyelidikan. Konsep asas dan pencirian laser serta aplikasinya turut dibincangkan.

This course covers the knowledge and advanced aspect of the field of optics leading to applications and research. Fundamental concept and characteristics of laser as well as its application are also discussed.

PHY4406 Gentian Optik dan Optoelektronik/ *Fiber Optics and Optoelectronics* 3(3+0)

Prasyarat : PHY3400

Kursus ini merangkumi konsep asas optoelektronik. Topik am di dalam optoelektronik termasuk pandu gelombang dielektrik, sumber dan pancaran cahaya, peranti cahaya dan pengesanan cahaya dibincangkan.

This course covers fundamental concepts in optoelectronics. General topics in optoelectronics including wave dielectric waveguides, light source and emission, light devices and light detectors are discussed.

- PHY4407 Teknologi Bahan Fotonik / *Photonics Material Technology* 3(3+0)
 Prasyarat : Tiada
 Kursus ini merangkumi prinsip umum interaksi gelombang cahaya dan sifat rambatan cahaya dalam bahan bagi pembangunan teknologi fotonik. Hukum penting dalam teori gelombang cahaya dibincangkan.
This course covers the general principles of light wave interaction and light propagation properties in materials for development of photonics technology. Important laws in light wave theory are discussed.
- PHY4502 Fizik Sinaran dan Radiobiologi/ *Radiation Physics and Radiobiology* 3(3+0)
 Prasyarat : PHY3105
 Kursus ini merangkumi pendedahan mengenai sumber, ciri dan saling tindak sinaran mengion dengan jirim. Peralatan pengesanan sinaran, kesan biologi, perlindungan dan aplikasi sinaran mengion turut ditekankan.
This course covers exposure to the source, properties and interactions of ionizing radiation with matter. Radiation detection equipment, biological effects, radiation protection and applications of ionizing radiation are also emphasized.
- PHY4503 Kerelatifan Khas dan Teori Medan Klasik/ *Special Relativity & Classical Field Theory* 3(3+0)
 Prasyarat : PHY3105
 Kursus ini merangkumi prinsip kerelatifan khas, struktur ruang-masa dan analisis tensor. Penggunaannya dalam kinematik, dinamik, keelektromagnetan dan medan klasik lain turut ditekankan.
This course covers the principles special relativity, the structure of space-time and tensor analysis. Their use in kinematics, dynamics, electromagnetism and other classical fields are also emphasized.
- PHY4505 Fizik Perubatan/ *Medical Physics* 3(3+0)
 Prasyarat : Tiada
 Kursus ini merangkumi pengetahuan dan kemahiran asas dalam fizik perubatan. Aspek dosimetri perubatan, prinsip radiologi, onkologi sinaran dan perubatan nuklear turut dibincangkan.
This course covers the basic knowledge and skills in medical physics. Aspect of medical dosimetry, the principle of radiology, radiation oncology and nuclear medicines are also discussed.
- PHY4506 Fizik Zarah dan Nuklear/ *Nuclear and Particle Physics* 3(3+0)
 Prasyarat : PHY3105
 Kursus ini merangkumi pengetahuan asas fizik zarah dan tenaga nuklear. Fenomenologi nuklear, model nukleus, ciri, tindak balas dan aplikasinya turut dibincangkan.
This course covers the basic knowledge of particle physics and nuclear energy. Nuclear phenomenology, nucleus models, properties, reaction and its applications are also discussed.
- PHY4600 Kaedah Matematik dalam Fizik II/ *Mathematical Methods in Physics II* 3(3+0)
 Prasyarat : PHY3600
 Kursus ini merangkumi teknik matematik lanjutan dalam penyelesaian masalah fizik. Struktur dan pencirian teknik bermatematik dan aplikasinya dalam fizik turut dibincangkan.
This course covers advanced mathematical techniques in solving physics problem. The structure and properties of mathematical techniques and its application in physics are also discussed.

PHY4603	Mekanik Kuantum Lanjutan/ <i>Advanced Quantum Mechanics</i>	3(3+0)
Prasyarat : PHY3601 Kursus ini merangkumi pengetahuan dan aspek lanjutan kaedah mekanik kuantum kepada sistem yang lebih realistik dan terperinci. Teori usikan, penghampiran dan sistem berbilang jasad turut dibincangkan. <i>This course covers the knowledge and advanced aspect of quantum mechanical methods to more realistic and detailed system. Perturbation, approximation theory and many body system are also discussed.</i>		
PHY4604	Maklumat dan Teknologi Kuantum/ <i>Quantum Information and Technology</i>	3(3+0)
Prasyarat : PHY3601 Kursus ini merangkumi konsep umum dalam maklumat dan teknologi kuantum. Unsur maklumat kuantum, komunikasi kuantum, komputasi kuantum dan realisasi teknologi kuantum turut ditekankan. <i>This course covers general concepts in quantum information and technology. The of elements of quantum information, quantum communication, quantum computing and realization of quantum technology are also emphasized.</i>		
PHY4605	Kaedah Pengkomputeran dalam Fizik/ <i>Computational Methods in Physics</i>	3(2+1)
Prasyarat : PHY3600 Kursus ini merangkumi kaedah berangka untuk penyelesaian dalam kamiran, pembezaan dan operasi matriks. Penggunaan teknik analisis berangka dalam fizik klasik dan kuantum juga dibincangkan. <i>This course covers numerical methods for solutions in integration, differentiation and matrix operation. The use of numerical analyses techniques in classical and quantum physics are also discussed.</i>		
PHY4701	Sains Data untuk Ahli Fizik dan Ahli Teknologi/ <i>Data Science for Physicist and Technologist</i>	3(3+0)
Prasyarat : PHY3600 Kursus ini merangkumi konsep pemahaman tugas pengiraan, manipulasi dan analisa data bagi penyelesaian masalah berkaitan fizik. Asas penyusunan dan peroperasian data dengan menggunakan program pengaturcaraan R serta pengenalan kepada konsep pembelajaran mesin dan dataraya dibincangkan. <i>This course covers the concept of understanding the role of computation, data manipulation and analysis for problem solving related to physics. The basics of data organization and operation by using R programming language as well as Introduction to the concept of machine learning and big data are discussed.</i>		
PHY4702	Rangkaian Neural Buatan/ <i>Artificial Neural Networks</i>	3(3+0)
Prasyarat : PHY3600 Kursus ini merangkumi prinsip kecerdasan buatan, teknik dan struktur pengaturcaraan pintar. Algoritma genetik, rangkaian perseptron dan logik kabur dibincangkan. <i>This course covers the basics of Artificial Intelligence, techniques and structures of intelligent programming. The genetic algorithms, perceptron networks and fuzzy logic are discussed.</i>		
PHY4900	Amali Lanjutan dalam Fizik/ <i>Advanced Practical in Physics</i>	1(0+1)
Prasyarat : PHY3105		

Kursus ini merangkumi amali lanjutan fizik dalam bidang fizik keadaan pepejal, fizik moden, optik, fizik terma dan keelektromagnetan. Pengendalian peralatan makmal dalam teknik sains eksperimen yang berkaitan turut ditekankan.

This course covers the advanced physics practical in the field of solid state physics, modern physics, optics, thermal physics and electromagnetism. The handling of laboratory equipment in related experimental science techniques are also emphasized.

PHY4902 Tajuk Khas/ *Special Topics*

3(3+0)

Prasyarat : PHY3105

Kursus ini merangkum tajuk tertentu/pilihan dalam bidang fizik pada peringkat tinggi. Kursus ini dirancang untuk memberikan kefahaman yang mendalam mengenai kajian khusus dan perkembangan terbaru dalam bidang fizik. Pemilihan topik akan ditentukan oleh jabatan.

This course covers encompasses selected topics at advanced level in physics. An indepth understanding of specialized fields in physics and/or recent advances in physics is discussed. The choice of topics is determined by the department.

PHY4903 Latihan Industri/ *Industrial Training*

8(0+8)

Prasyarat : Dengan kebenaran jabatan

Kursus ini merangkumi pendedahan kepada persekitaran kerja yang sebenar di industri/organisasi. Aplikasi teori, praktikal, kemahiran komunikasi dan penyelesaian masalah turut ditekankan.

This course covers the exposure to real working environment in industries/organizations. Application of the theoretical, practical, communication and problem solving skills are also emphasized.

PHY4959 Disertasi Bachelo/ *Bachelor Dissertation*

6(0+6)

Prasyarat : Dengan kebenaran jabatan

Kursus ini merangkumi penyediaan cadangan, pelaksanaan dan penulisan saintifik untuk sesuatu projek penyelidikan. Pendekatan saintifik bagi menjana data secara sistematik melalui reka bentuk, pengumpulan dan analisis data yang sesuai turut ditekankan.

This course covers the preparation of proposal, implementation and scientific writing of research project. Scientific approach to generate data systematically through appropriate design, data collection and analysis are also emphasized.)

SINOPSIS KURSUS / COURSE SYNOPSIS

Jabatan Kimia/ Department of Chemistry

CHY2001 Kimia Am/ General Chemistry

4(3+1)

Prasyarat : Tiada/ None

Kursus ini membincangkan tajuk umum dalam bidang kimia fizik, tak organik dan organik meliputi struktur atom dan molekul, ikatan kimia, keseimbangan dan kinetik tindak balas serta kimia logam. Penamaan, sifat dan tindak balas beberapa kumpulan berfungsi sebatian organik turut dibincangkan.

This course discusses general topics in physical, inorganic and organic chemistry including atomic and molecular structure, chemical bonding, reaction's equilibrium and kinetics, and chemistry of metals. Nomenclature, properties, and reactions of various functional groups of organic compounds are also discussed.

CHY3001 Prinsip Kimia/ Principle of Chemistry

4(3+1)

Prasyarat : Tiada/ None

Kursus ini merangkumi konsep dan aplikasi teori prinsip asas kimia. Ini termasuklah prinsip asas dalam pengukuran dan pengiraan dalam bidang kimia fizik, kimia organik dan kimia tak organik, serta menggunakan konsep asas ini dalam bidang kimia polimer dan kimia industri.

This course covers basic concepts and applications of chemistry. This includes basic concepts in measurement and calculation in the field of physical chemistry, organic chemistry and inorganic chemistry, as well as applying these concepts in the field of polymer chemistry and industrial chemistry.

CHY3002 Kimia Fizik dan Tak Organik/ Physical and Inorganic Chemistry

4(3+1)

Prasyarat : Tiada/ None

Kursus ini merangkumi aspek asas dalam kimia fizik dan tak organik. Ini termasuk teori atom moden, jadual berkala dan sifat berkala, kimia kumpulan utama, teori pengikatan, sifat gas, cecair dan pepejal, keseimbangan kimia, elektrokimia, termodinamik, kinetik dan kimia nuklear.

This course covers basic aspects of physical and inorganic chemistry. These include modern atomic theory, periodic table and periodic properties, main group element, theory of bonding, properties of gas, liquid and solid, chemical equilibrium, electrochemistry, thermodynamics, kinetics and nuclear chemistry.

CHY3003 Asas Keselamatan Makmal/ Basic Laboratory Safety

2(2+0)

Prasyarat : Tiada/ None

Kursus ini merangkumi aspek pengurusan keselamatan dan amalan kerja yang baik semasa di dalam makmal serta peraturan yang berkaitan. Pengendalian bahan kimia dan sisa bahan kimia serta bahaya (hazard) makmal juga dibincangkan.

This course covers aspects of safety management and good work practices while in the laboratory and related regulations. Handling of chemicals and chemical waste as well as laboratory hazards are also discussed.

CHY3004 Pembelajaran Perkhidmatan dalam Kimia/ Service Learning in Chemistry

1(0+1)

Prasyarat : Tiada/ None

Kursus ini merangkumi aktiviti bagi meningkatkan kesedaran dan kefahaman komuniti tentang

kepentingan bidang kimia dalam kehidupan harian. Pengetahuan konsep kimia diaplikasikan melalui aktiviti kerjasama dengan masyarakat luar atau industri.

This course encompasses activities to increase awareness and understanding of the community on the importance of chemistry in daily lives. The knowledge of concept in chemistry is applied through joined activities with the community or industry.

CHY3101 Kimia Fizik I/ *Physical Chemistry I* 4(3+1)

Prasyarat : Tiada/ *None*

Kursus ini merangkumi konsep-konsep penting bidang kimia fizik seperti teori dan aplikasi termodinamik, kinetik, elektrokimia, dan teori kuantum. Keseimbangan kimia dan pengiraan asas mekanisme tindak balas juga dibincangkan.

This course covers the important concepts in physical chemistry such as the theory and application of thermodynamics, kinetics, electrochemistry, and quantum theory. Chemical equilibrium and basic calculation of reaction mechanism are also discussed.

CHY3102 Kimia Fizik II/ *Physical Chemistry II* 4(3+1)

Prasyarat : CHY3101

Kursus ini merangkumi aspek kimia fizik berkaitan dengan teori kinetik, mekanisme tindak balas dan tindak balas kompleks. Hukum termodinamik, larutan, keseimbangan fasa, elektrolit dan elektrokimia, koloid dan kuantum kimia turut dibincangkan.

This course covers aspects of physical chemistry related to kinetic theory, reaction mechanism and complex reactions. Thermodynamic laws, solutions, phase equilibria, electrolytes, electrochemistry, colloids and quantum chemistry are also discussed.

CHY3103 Kimia Polimer/ *Polymer Chemistry* 4(3+1)

Prasyarat : CHY3101

Kursus ini merangkumi aspek kimia polimer berkaitan dengan jenis, ciri dan sifat polimer serta mekanisme tindak balas dan kinetik pempolimeran. Eksperimen melibatkan sintesis dan pencirian polimer dilaksanakan.

This course covers aspects of polymer chemistry related to types, characteristics and properties of polymers as well as mechanism of reactions and kinetics of polymerization. Experiments involving synthesis and characterization of polymers are performed.

CHY3201 Kimia Organik I/ *Organic Chemistry I* 4(3+1)

Prasyarat : CHY2001 atau/or setara/ *equivalent*

Kursus ini merangkumi topik asas dalam kimia organik seperti struktur, ikatan, tatanama, keisomeran geometri dan optik, sifat, tindak balas, dan sintesis pelbagai kelas sebatian organik. Eksperimen ke atas kumpulan berfungsi utama sebatian organik dijalankan.

This course covers basic topics in organic chemistry such as structure, bonding, nomenclature, geometrical and optical isomerism, properties, reactions, and synthesis of various classes of organic compounds. Experiments on the main functional groups of organic compounds are carried out.

CHY3202 Kimia Organik II/ *Organic Chemistry II* 4(3+1)

Prasyarat : CHY3201

Kursus ini merangkumi kimia sebatian aromatik dan heterosiklik, tindakbalas enolat, kawalan stereokimia dalam tindak balas penukargantian dan penyediaan alkena, prinsip asas kimia hijau dan kimia biomolekul. Penyediaan sebatian kimia menggunakan kaedah sintesis organik terpilih dijalankan.

This course covers the chemistry of aromatic and heterocyclic compounds, reactions of enolates, stereochemical control during substitution reactions and preparation of alkenes, basic principle of green chemistry and chemistry of biomolecules. Preparation of chemical compounds using selected synthetic organic methods are carried out.

CHY3203 Sintesis Organik / Organic Synthesis 4(3+1)

Prasyarat : CHY3202 dan/ and CHY3402

Kursus ini merangkumi topik yang berkaitan dengan strategi sintetik dalam kimia organik, sintesis tak simetri serta tindak balas pengoksidaan, penurunan, perisiklik, radikal, fotokimia dan organologam termasuk pengenalan ringkas kepada kimia perubatan. Sintesis pelbagai langkah sebatian kimia turut dijalankan.

This course covers topics related to synthetic strategies in organic chemistry, asymmetric synthesis oxidation as well as reduction, pericyclic, radical, photochemical, and organometallic reactions including brief introduction on medicinal chemistry. Multi-step syntheses of organic compounds are conducted.

CHY3204 Kimia Hasil Semula Jadi/ Natural Products Chemistry 4(3+1)

Prasyarat : CHY3202 dan/ and CHY3402

Kursus ini memberikan gambaran menyeluruh metabolit primer dan kelas utama metabolit sekunder, biosintesis, pemencilan menggunakan teknik kromatografi dan analisis struktur hasil semula jadi dengan kaedah spektroskopi maju. Aplikasi hasil semula jadi disertakan bagi memberi penghargaan mengenai kepentingannya kepada manusia.

This course presents an overview of the primary metabolites and the major classes of secondary metabolites, biosynthesis, isolation using chromatographic techniques and structural analysis by advanced spectroscopic methods. The applications of natural products are included to give appreciation of their importance to humankind.

CHY3205 Kimia Protein/ Protein Chemistry 3(3+0)

Prasyarat : CHY3202 dan/ and CHY3402

Kursus ini merangkumi aspek kimia bagi protein, peptida dan nukleotida. Ia meliputi pengenalan kepada asid amino dan nukleotida, struktur, sintesis dan pencirian biofizikal protein dan peptida, protein termodinamik, tindak balas berenzim, ubatan berasaskan protein serta topik khas dalam kejuruteraan protein.

This course covers the chemistry aspects of protein, peptide and nucleotides. It includes an introduction to amino acids and nucleotides, structures, synthesis and biophysical characterization of protein and peptide, protein thermodynamics, enzymatic reactions, protein-based drug, and special topics in protein engineering.

CHY3301 Keberkalaan dan Ikatan Kimia/ Periodicity and Chemical Bonding 4(3+1)

Prasyarat : CHY2001 atau/or setara/ equivalent

Kursus ini membincangkan beberapa aspek asas kimia tak organik yang merangkumi jadual berkala dan sifatnya. Unsur kumpulan utama, unsur peralihan, struktur molekul, ikatan, dan interaksi antara molekul juga dibincangkan.

This course discusses several basic aspects of inorganic chemistry which covers periodic table and its properties. The main group elements, transition elements, molecular structure, bonding, and intermolecular interactions are also discussed.

CHY3302 Teori Kumpulan dan Struktur Pepejal/ Group Theory and Solid Structure 3(2+1)

Prasyarat : CHY3301

Kursus ini merangkumi, simetri molekul dan kumpulan titik serta penggunaan teori kumpulan dan prinsip simetri dalam ikatan dan spektroskopi getaran molekul. Susunan atom dan kecacatan hablur dalam pepejal tak organik terpilih turut dibincangkan.

This course covers molecular symmetry, point group including the use of group theory and symmetry principles in bonding and vibrational spectroscopy of molecules. Atomic arrangements and crystal defects in selected inorganic solids are also discussed.

CHY3303 Kimia Koordinatan dan Organologam Logam Peralihan/ *Coordination and Organometallic Chemistry of Transition Metals* 3(2+1)

Prasyarat : CHY3302

Kursus ini merangkumi kimia koordinatan logam peralihan yang merangkumi struktur, teori ikatan, spektrum elektron, kemagnetan, dan jenis tindak balas. Ia juga merangkumi kimia organologam blok d dengan aplikasinya dalam pemangkinan homogen terpilih.

This course covers transition metals coordination chemistry which includes structure, bonding theories, electronic spectrum, magnetism and type of reactions. It also includes d-block organometallic chemistry with its application in selected homogeneous catalysis.

CHY3304 Struktur Pepejal dan Kimia Koordinatan/ *Solid Structure and Coordination Chemistry* 4(3+1)

Prasyarat : CHY3301

Kursus ini merangkumi, simetri molekul, teori kumpulan, struktur hablur dan kecacatan hablur dalam pepejal tak organik. Ia juga berkaitan dengan kimia koordinatan logam peralihan yang merangkumi struktur, teori ikatan, spektrum elektron, kemagnetan, jenis tindak balas serta kimia organologam.

This course covers molecular symmetry, group theory, crystal structure and crystal defects of inorganic solids. It also covers transition metals coordination chemistry which includes structure, bonding theories, electronic spectrum, magnetism, type of reactions as well as organometallic chemistry.

CHY3401 Kimia Analisis/ *Analytical Chemistry* 3(2+1)

Prasyarat : CHY3101 atau/ or CHY3002

Kursus ini merangkumi kaedah asas dan amali dalam kimia analisis. Ini termasuk konsep asas dalam analisis data yang digunakan dalam pengukuran seperti kaedah titrimetri, gravimetri dan pengekstrakan pelarut. Analisis instrumentasi seperti teknik kromatografi, elektrokimia dan analisis terma turut dibincangkan

This course covers the basic methods and practical in analytical chemistry. This includes basic concept in data analysis used in measurements such as titrimetry, gravimetry, solvent extraction methods. Instrumentation analysis such as chromatography, electrochemistry and thermal analysis techniques are also discussed.

CHY3402 Spektroskopi Kimia/ *Chemical Spectroscopy* 4(3+1)

Prasyarat : CHY3101 dan/ and CHY3201

Kursus ini merangkumi pengenalan kepada kaedah spektroskopi dalam bidang analisis kimia. Data spektroskopi ditafsirkan secara kuantitatif dan kualitatif. Pendedahan potensi perniagaan dijalankan secara kajian kes.

This course covers the introduction to spectroscopic methods in chemical analysis. Spectroscopic data is interpreted quantitatively and qualitatively. Exposure to business potential is conducted through

case studies.

CHY3501 Kimia Perindustrian/ *Industrial Chemistry* 3(3+0)

Prasyarat : CHY3202

Kursus ini merangkumi pengenalan dan huraian berkaitan prinsip pemprosesan sumber-sumber asli yang terdiri daripada bahan organik dan tak organik yang digunakan dalam industri pemprosesan bahan kimia. Permasalahan dan cabaran dalam sektor pembersihan, pembuatan dan polimer sintetik turut dibincangkan.

This course covers the introduction and description of the principle of natural resource processing that derives from organic and inorganic compounds used in the chemical processing industries. The problems and challenges in the cleaning, manufacturing, and synthetic polymer sectors are also discussed.

CHY3502 Kimia Organik Perindustrian/ *Industrial Organic Chemistry* 3(3+0)

Prasyarat : CHY3201

Kursus ini merangkumi aspek perindustrian berasaskan bahan kimia organik. Proses penghasilan, penggunaan dan kepentingan bahan kimia organik dibincangkan.

This course covers aspects of industrial organic chemistry. The manufacturing processes, applications and importance of organic chemical products are discussed.

CHY3503 Kimia Tak Organik Perindustrian/ *Industrial Inorganic Chemistry* 3(3+0)

Prasyarat : CHY3304

Kursus ini merangkumi aspek perindustrian yang berkaitan dengan kimia tak organik. Proses penghasilan dan kegunaan produk serta bahan termaju dalam industri kimia tak organik dibincangkan.

This course covers industrial aspects related to inorganic chemistry. The manufacturing processes and product applications, as well as advanced materials in the inorganic-based industries are discussed.

CHY3504 Prinsip Teknologi Kimia/ *Chemical Technology Principles* 4(4+0)

Prasyarat : CHY3102

Kursus ini merangkumi proses dan operasi unit dalam industri kimia. Faktor yang terlibat dalam kejuruteraan kimia serta kawalan proses ditekankan melalui pendekatan kimia. Masalah dan kawalan berkaitan proses dan teknologi kimia juga dibincangkan.

This course covers processes and unit operations in chemical industry. Factors that are involved in chemical engineering and process controls are highlighted through chemistry approach. Problems and controls that are related to processes and chemical technology are discussed.

CHY3505 Plastik dan Komposit/ *Plastics and Composites* 3(3+0)

Prasyarat : CHY3103

Kursus ini merangkumi teknologi pemprosesan dan pengeluaran produk berasaskan plastik dan komposit dalam industri dan juga kegunaannya dalam teknologi termaju termasuk bioperubatan dan elektronik. Pembangunan bahan boleh diperbaharui iaitu bioplastik dan biokomposit juga dibincangkan.

This course covers the technology of processing and production of plastic and composite-based products in the industry as well as the applications in advanced technologies including biomedical and electronics. The development of renewable materials which are bioplastics and biocomposites is

discussed.

CHY3506 Oleokimia/ Oleochemistry

3(2+1)

Prasyarat : CHY3202

Kursus ini merangkumi prinsip asas, proses kimia serta analisis berkaitan minyak dan lemak termasuk terbitan oleokimia. Penggunaan teknologi terkini dalam pemprosesan minyak dan lemak termasuk mikroemulsi dan bioteknologi juga dibincangkan. Pendedahan kepada potensi perniagaan dijalankan secara kajian kes.

This course covers basic principle, chemical process and analyses related to oil and fats, including oleochemical derivatives. The use of the latest technology in oil and fats processing including microemulsion and biotechnology is also discussed. Exposure to business potential is conducted through case studies.

CHY3507 Kimia dan Teknologi Getah/ Rubber Chemistry and Technology

2(2+0)

Prasyarat : CHY3103

Kursus ini merangkumi teknologi dan kimia getah asli dan sintetik. Bidang tumpuan adalah asas kimia getah asli dan sintetik, pemprosesan dan sistem pemvulkanan serta analisis getah dan produk getah. Teknologi dan kegunaan getah dalam industri dan komuniti serta masalah dalam industri getah dan penyelesaiannya dibincangkan.

This course covers the technology and chemistry of natural and synthetic rubbers. Areas of focus are the foundation of the chemistry of natural and synthetic rubbers, the processing and vulcanization systems as well as the analysis of rubber and rubber products. The technology and application of rubbers in industries and communities, as well as problems in the rubber industry and their solutions are discussed.

CHY3508 Kimia Hijau/ Green Chemistry

3(2+1)

Prasyarat : CHY3402

Kursus ini merangkumi pelbagai teknik dan teknologi kimia hijau yang digunakan dalam penyelidikan dan pembangunan sektor industri. Eksperimen menggunakan kaedah kimia hijau turut dijalankan.

This course covers various techniques and technologies in green chemistry that are used in research and development in the industrial sector. Experiments using green chemistry techniques are also conducted.

CHY3601 Kimia Petroleum/ Petroleum Chemistry

3(3+0)

Prasyarat : CHY3202

Kursus ini merangkumi aspek pembentukan sehingga penghasilan petroleum. Kaedah eksplorasi dan penggerudian minyak dan gas asli dibincangkan. Isu terkini dalam kimia petroleum turut diberi penekanan.

This course covers various aspect of petroleum formation to production. The exploration method and drilling of oil and gas are discussed. Current issues in petroleum chemistry are also emphasized.

CHY3602 Proses Penapisan Petroleum/ Petroleum Refining Processes

3(3+0)

Prasyarat : CHY3202

Kursus ini merangkumi pengenalan kepada proses penyulingan petroleum dan bahan hasil penapisan. Kaedah penyulingan minyak mentah, proses pengkokan tertunda dan proses penapisan turut dibincangkan.

This course covers an introduction to the petroleum refining process and refinery products. Crude distillation method, delayed coking and refinery process are also discussed.

CHY3603 Petrokimia dari Hidrokarbon Ringan/ *Petrochemicals from Light Hydrocarbon* 3(3+0)

Prasyarat : CHY3602

Kursus ini merangkumi pelbagai proses penghasilan bahan kimia perantara dan produk siap yang berasal dari hidrokarbon ringan C1-C4. Keadaan tindak balas dan carta aliran proses juga dibincangkan

This course covers various processes for production of chemicals intermediates and finished products derived from light hydrocarbon of C1-C4. Reaction conditions and flow chart of the processes are also discussed.

CHY3604 Petrokimia dari Sebatian Aromatik/ *Petrochemicals from Aromatics Compounds* 3(3+0)

Prasyarat : CHY3603

Kursus ini merangkumi pelbagai proses penghasilan sebatian aromatik dan bahan kimia terbitannya. Keadaan tindak balas dan carta aliran proses juga dibincangkan.

Kursus ini merangkumi pelbagai proses penghasilan sebatian aromatik dan bahan kimia terbitannya. Keadaan tindak balas dan carta aliran proses juga dibincangkan.

CHY3605 Kawalan Tumpahan Minyak/ *Oil Spill Control* 3(3+0)

Prasyarat : CHY3601

Kursus ini merangkumi aspek tumpahan minyak, kaedah pencegahan dan kawalannya. Ciri dan komposisi minyak, sifat tumpahan, kesan tumpahan, kaedah pelupusan serta peraturan antarabangsa yang berkaitan tumpahan minyak turut dibincangkan.

This course covers aspects in oil spill, methods of prevention and its control. Type and composition of the oil, the nature of the spill, the impact of the spill, disposal method and the relevant international regulations related to oil spill are also discussed.

CHY3606 Amali dalam Kimia Petroleum/ *Practical in Petroleum Chemistry* 1(0+1)

Prasyarat : CHY3603

Kursus ini merangkumi pelbagai aspek dalam pengendalian eksperimen dalam kimia petroleum yang berkait dengan industri minyak dan gas. Analisis bahan di dalam air efluen turut dijalankan.

This course covers various aspects in handling petroleum chemistry experiment that are related in oil and gas industry. The material analysis in effluent water is also performed..

CHY4101 Kinetik Kimia/ *Chemical Kinetics* 3(2+1)

Prasyarat : CHY3102

Kursus ini merangkumi tajuk berkaitan dengan kinetik kimia dalam fasa gas dan cecair, tindak balas bermangkin asid bes dan enzim, jerapan, tindak balas permukaan dan cepat. Terdapat juga penekanan pada aplikasi kimia kinetik dan pemangkinan melalui eksperimen.

This course covers topics related to chemical kinetics in the gases and liquid phases, acid-base catalysis and enzymatic reactions, adsorption, surface and fast reactions. There is also emphasis on applications of chemical kinetics and catalysis through experiments.

CHY4102 Termodinamik Kimia/ *Chemical Thermodynamics* 3(2+1)

Prasyarat : CHY3102

Kursus ini meliputi perbincangan tentang transformasi tenaga bermula daripada unit terkecil iaitu atom dan molekul sehinggalah kepada jirim pukal merangkumi kuantiti, penyerakan, hubungan kait di antara bentuk tenaga yang berbeza serta pengaruhnya terhadap proses kimia dan kestabilan fasa bahan dan larutan. Termodinamik berkaitan penjanaan dan storan tenaga diperbaharui juga dibincangkan.

This course covers the discussion on energy transformation from individual atoms and molecules to bulk matter encompassing quantity, dispersal, interrelationship of different forms of energy and their influence on chemical processes and the stability of substances and solutions. Thermodynamics in relation to renewable energy and storage are also included in the discussion.

CHY4103 Kimia Pengkomputeran/ *Computational Chemistry* 4(3+1)

Prasyarat : CHY3102

Kursus ini merangkumi aspek asas pemodelan molekul seperti mekanik klasik dan mekanik kuantum. Aspek teori dan gunaan yang melibatkan kajian dinamik molekul dan pengiraan fungsi gelombang zarah turut dibincangkan.

This course covers fundamental aspects of molecular modeling such as classical mechanics and quantum mechanics. Theoretical and practical aspects involving the study of molecular dynamics and the calculation of particle wave functions are also discussed.

CHY4201 Tajuk Khas dalam Kimia Organik/ *Special Topics in Organic Chemistry* 3(3+0)

Prasyarat : CHY3203 atau/ or CHY3204

Tajuk terpilih dalam kimia organik lanjutan ditentukan oleh Jabatan akan dibincangkan dengan lebih mendalam.

Selected topics in advanced organic chemistry determined by the Department will be discussed.

CHY4301 Kimia Keadaan Pepejal/ *Solid State Chemistry* 3(3+0)

Prasyarat : CHY3302 atau/ or CHY3304

Kursus ini merangkumi kaedah sintesis, komposisi dan struktur bagi pelbagai pepejal yang mempunyai fungsi yang berlainan. Pemahaman lanjutan terhadap korelasi antara sifat elektrik, magnet dan optik juga dibincangkan

This course covers synthesis methods, compositions and structures of various solids with different functions. An advanced understanding of the correlation between electrical, magnetic and optical properties is also discussed.

CHY4302 Kimia Biotakorganik/ *Bioinorganic Chemistry* 3(3+0)

Prasyarat : CHY3303 atau/ or CHY3304

Kursus ini memberi tumpuan kepada penggunaan ubat-ubatan berasaskan logam untuk merawat dan mengesan penyakit: sintesis, struktur, sifat umum, mekanisme yang diketahui dan hubungan struktur-aktiviti. Ia juga merangkumi reka bentuk supramolekul, swapemasangan dan aplikasi biologi.

This course focuses on the usage of metallo-drugs for treating and detecting disease, synthesis, structure, general properties, known mechanism and structure-activity relationship. It also includes supramolecular architecture, self-assembly and their biological applications.

CHY4401 Elektrokimia/ *Electrochemistry* 3(3+0)

Prasyarat : CHY3101 dan/ and CHY3401

Kursus ini merangkumi prinsip elektrokimia seperti aktiviti ion, kekonduksian, lapisan berganda elektrik, proses elektrod, voltametri, potensiometri dan pengelektroenan. Kaedah penyelesaian masalah elektrokimia turut dibincang.

This course covers the principles of electrochemistry such as ion activity, ionic mobility, electrical double layer, electrode processes, voltammetry and electrodeposition). The solutions of electrochemical problems also are also discussed.

CHY4402 Teknik Pemisahan/ *Separation Techniques* 3(3+0)

Prasyarat : CHY3401

Kursus ini merangkumi pengenalan kepada prinsip asas kaedah pemisahan, kaedah pemisahan klasik dan terkini. Penganalisaan data daripada kaedah pemisahan dalam pelbagai aplikasi dijalankan.

This course covers the introduction to the basic principle of separation methods, the classical and advance separation methods. Analysis of data from separation methods in various applications is performed.

CHY4403 Kimia Forensik/ *Forensic Chemistry* 3(3+0)

Prasyarat : CHY3402

Kursus ini merangkumi kimia analisis sains forensik seperti jaminan kualiti/kawalan kualiti, dadah, toksikologi, analisis jejak dan pembakaran, dan serologi. Beberapa kaedah analisis forensik dirangkumi termasuk teknik kromatografi dan spektroskopi.

This course covers the analytical chemistry of forensic science such as quality assurance/quality control, drug, toxicology, trace and arson analysis, and serology. Several methods of forensic analysis are covered including chromatography and spectroscopy techniques.

CHY4501 Keusahawanan Kimia/ *Chempreneurship* 3(3+0)

Prasyarat : CHY3501 atau/ or CHY3504

Kursus ini merangkumi aspek perniagaan dalam kimia bermula dengan penemuan dalam makmal, langkah pengeluaran penemuan dalam industri kepada tata tertib bidang kimia. Kajian kes meliputi pembentangan cadangan perniagaan dalam bidang kimia dan produk pengguna.

This course covers aspects of business in chemistry, commencing from creation in the lab, steps for introducing invention in the industry to regulations in the field of chemistry. The case study comprises pitching business proposals in the field of chemistry and consumer products.

CHY4601 Pemangkinan heterogen/ *Heterogeneous Catalysis* 3(2+1)

Prasyarat : CHY3102

Kursus ini merangkumi pengenalan kepada pemangkinan heterogen dan pengklasifikasiannya, teknik penyediaan dan pencirian mangkin serta tindak balas bermangkin. Aplikasi mangkin dalam industri dan pengawalan pencemaran alam sekitar juga turut dibincangkan.

This course covers introduction to the heterogeneous catalyst and its classification, catalyst preparation and characterization techniques and catalytic reaction. The application of catalysts in industries and environmental pollution control also also discussed.

CHY4903 Latihan Industri/ *Industrial Training* 8(0+8)

Prasyarat : Dengan Kebenaran Jabatan/ *With Departmental Approval*

Kursus ini merangkumi pendedahan kepada persekitaran kerja yang sebenar di industri/organisasi. Aplikasi teori, praktikal, kemahiran komunikasi dan penyelesaian masalah turut ditekankan.

This course covers the exposure to real working environment in industries/organizations. Application of the theoretical, practical, communication and problem-solving skills are also emphasized.

CHY4959 Disertasi Bachelo/ *Bachelor Dissertation*

6(0+6)

Prasyarat : Dengan Kebenaran Jabatan/ With Departmental Approval

Kursus ini merangkumi kajian literatur, penggunaan teknik penyelidikan yang sesuai, pengumpulan dan analisis data. Penekanan turut diberikan kepada penafsiran keputusan, perbincangan dan kesimpulan kajian saintifik dalam projek penyelidikan.

This course covers literature reviews, use of appropriate research techniques, data collections and analyses. Emphases are also given to the interpretation of results, discussions and conclusions of scientific studies in research projects.

**SINOPSIS KURSUS, JABATAN MATEMATIK DAN STATISTIK /
COURSE SYNOPSIS, DEPARTMENT OF MATHEMATICS AND STATISTICS**

**(i) Bacelor Sains Matematik dengan Pendidikan (Kepujian)/
Bachelor of Science in Mathematics with Education (Honours)**

MTH3100 Kalkulus/ *Calculus* 3(3+0)

Prasyarat : Tiada

Kursus ini merangkum pembinaan konsep dalam kalkulus satu pembolehubah, konsep set dan fungsi untuk memahami idea mengenai had, kesinjaran dan terbitan. Pembezaan dan teorem yang berkaitan kamiran sebagai proses anti-terbitan serta teknik kamiran ditekankan.

This course covers the building up of the concepts in calculus of one variable, the concept of sets and functions to understand the idea of limits, continuity and derivatives. Differentiations and theorems related to integration as a process of anti-derivatives together with the integration techniques are emphasized.

MTH3101 Kalkulus Lanjutan/ *Advanced Calculus* 3(3+0)

Prasyarat : MTH3100

Kursus ini merangkum teorem asas kalkulus, diikuti dengan fungsi banyak pembolehubah yang melibatkan kalkulus pembezaan dan kamiran. Jujukan dan siri nombor nyata dan fungsi nyata turut dibincangkan.

This course covers theorems on elementary calculus, followed by functions of several variables involving differential and integral calculus. Sequence and series of real numbers and functions are also discussed.

MTH3102 Persamaan Pembezaan/ *Differential Equations* 3(3+0)

Prasyarat : MTH3100 dan MTH3200

Kursus ini merangkum pengelasan persamaan pembezaan dan kaedah penyelesaian persamaan pembezaan linear. Diikuti dengan kaedah membina penyelesaian am daripada beberapa penyelesaian khusus yang diperolehi, terutama dari satu set penyelesaian yang tidak bersandar linear. Kaedah pekali tak ditentukan dan ubahan parameter, jelmaan Laplace dan penggunaannya kepada masalah nilai awal dan nilai sempadan dibincangkan.

This course covers classification of differential equations and methods of solving linear differential equations. Followed by methods of constructing general solutions from several particular solutions obtained, especially from a set of linearly independent solutions. Methods of undetermined coefficients and variations of parameter, Laplace transform and its applications to initial value and boundary value problems are discussed.

MTH3103 Analisis Vektor/ *Vector Analysis* 3(3+0)

Prasyarat : MTH3100

Kursus ini merangkum aspek vektor dalam ruang berdimensi n , ($n > 2$), hasil darab titik dan hasil darab silang. Kaedah pembezaan vektor, kamiran vektor dan koordinat lengkung linear dibincangkan.

This course covers aspects of vectors in n -dimensional space, ($n > 2$), dot and cross products. Vector differentiation, vector integration and curvilinear coordinates are discussed.

MTH3104 Kaedah Matematik/ *Mathematical Methods* 3(3+0)

Prasyarat : MTH3102 dan MTH3201

Kursus ini merangkum konsep operasi linear, pengoperasi linear, perwakilan matriksnya dan siri Fourier. Penyelesaian bersiri bagi persamaan pembezaan biasa dan fungsi khas yang terjana,

persamaan pembezaan separa dan kaedah penyelesaian yang merangkum pemisah pembolehubah dan kaedah jelmaan dibincangkan.

The course covers the concept of linear operation, linear operators, their matrix representations and Fourier series. Series solutions to ordinary differential equations and the special functions generated, partial differential equations and methods of solutions which cover separable variable and transformation methods are discussed.

MTH3200 Aljabar/ Algebra

3(3+0)

Prasyarat : Tiada

Kursus ini merangkum teori mantik dan set, hubungan, fungsi, sistem nombor nyata dan kompleks, jujukan dan siri mudah, polinomial dan teori persamaan. Geometri koordinat, aljabar vektor dan penyelesaian sistem persamaan linear, asas dan sistem koordinat dalam R^2 dan R^3 dibincangkan.

This course covers mantic and set theories, relations, functions, real and complex number systems, elementary sequence and series, polynomials and theory of equations. Coordinate geometry, vector algebra and solutions to system of linear, basis and coordinate systems in R^2 and R^3 are discussed.

MTH3201 Aljabar Linear/ Linear Algebra

3(3+0)

Prasyarat : MTH3100 dan MTH3200

Kursus ini merangkum ruang vektor, asas bagi ruang vektor, transformasi linear, perwakilan matriks, pangkat and kenolan, nilai eigen, vektor eigen dan ruang eigen.

This course covers vector spaces, bases for vector space, linear transformation, matrix representation, rank and nullity, eigenvalues, eigenvectors and eigenspace.

MTH3202 Pengenalan Kepada Aljabar Moden/ Introduction to Modern Algebra

3(3+0)

Prasyarat : MTH3201

Kursus ini merangkum konsep kebolehbahagian, fungsi dan set integer. Ini diikuti dengan kekongruenan linear, hubungan kesetaraan, kumpulan, gelanggang, medan serta pemetaan. Ide asas mengenai hasil darab terus kumpulan, teori unggulan dan operasi asas melibatkan unggulan turut dibincangkan.

This course covers concepts related to divisibility, functions and the set of integers. This is followed by linear congruence, equivalence relations, group, rings, fields and mapping. Basic ideas on direct products of groups, theory of ideals and basic operation involving ideals are also discussed.

MTH3301 Analisis Nyata/ Real Analysis

3(3+0)

Prasyarat : MTH3101

Kursus ini merangkum konsep jujukan nombor nyata dan jenisnya, siri nombor serta ujian siri menumpu, ruang metrik, fungsi selanjar dan ide mengenai set terbuka dan tertutup dalam ruang tersebut. Ciri ruang metrik, teorem titik tetap dan teorem Heine-borel dan jenis ruang metrik dibincangkan, diikuti dengan perbincangan kamiran jenis Rieman dan Lesbegue.

This course covers the concept of sequence of real numbers and its types, number series and convergence tests, metric space, continuous functions and ideas concerning open and closed sets in such space. Characteristics of metric space, fixed point theorem and Heine-Borel theorem and types of metric spaces are discussed, followed by discussion on Rieman and Lesbegue integrals.

MTH3302 Analisis Kompleks/ Complex Analysis

3(3+0)

Prasyarat : MTH3101

Kursus ini merangkum aljabar nombor kompleks, fungsi analisis, fungsi permulaan dan

pemetaannya, kamiran kompleks, teorem dan rumus kamiran Cauchy. Teorem Liouville, teorem modulus maksimum, teorem asas aljabar, siri kuasa, siri Taylor, pensifar dan kutub, reja, teorem reja, penilaian kamiran kontur dan pemetaan mensebentuk turut dibincangkan.

The course covers algebra of complex numbers, analytic functions, elementary functions and mapping, complex integration, Cauchy's theorem and integration formula, Liouville's theorem, maximum modulus theorem, fundamental theorem of algebra, power series, Taylor's series, zeroes and poles, residues, the residue theorem, evaluation of contour integrals and conformal mapping are also discussed.

MTH3401 Kebarangkalian dan Statistik I/ *Probability and Statistics I* 3(3+0)

Prasyarat : MTH3100

Kursus ini merangkum konsep asas statistik termasuk pembolehubah rawak, kebarangkalian, taburan khas, jangkaan dan momen, penganggaraan dan ujian hipotesis, regresi dan korelasi.

This course covers fundamental concepts of statistics including random variables, probability, special distributions, expectations and moments, estimation and hypothesis testing, regression and correlation.

MTH3402 Kebarangkalian dan Statistik II/ *Probability and Statistics II* 3(3+0)

Prasyarat : MTH3401

Kursus ini merangkum konsep kebarangkalian, pembolehubah rawak, fungsi ketumpatan kebarangkalian, taburan kebarangkalian, jangkaan matematik dan fungsi penjana momen. Jenis taburan, fungsi pembolehubah rawak, penjelmaan pembolehubah dan ujian hipotesis dibincangkan.

This course covers probability concept, random variables, probability density functions, probability distributions, mathematical expectations and moment generating functions. Types of distributions, function of random variables, variable transformation and testing of hypothesis are discussed.

MTH3403 Rekabentuk Ujikaji/ *Experimental Design* 3(3+0)

Prasyarat : MTH3401

Kursus ini merangkum konsep asas statistik, prinsip asas rekabentuk ujikaji, matlamat serta penerapannya. Beberapa rekabentuk yang penting berserta dengan analisis yang bersesuaian dibincangkan. Rekabentuk yang dipertimbangkan ialah rekabentuk rawak lengkap, rekabentuk blok rawak lengkap, rekabentuk segiempat sama Latin, rekabentuk blok rawak tak lengkap seimbang, rekabentuk faktor dua dan tiga faktor dan rekabentuk faktor 2^k bagi $k=2$ dan $k=3$.

This course covers basic concepts in statistics, basic principles in design of experiments, their aims and implementations. Several important designs and their appropriate analyses are discussed. The designs considered are completely randomized design, randomized complete block design, Latin square design, balanced incomplete block design, two and three factors factorial design and 2^k factorial design for $k=2$ and $k=3$.

MTH3404 Model Linear/ *Linear Model* 3(3+0)

Prasyarat : MTH3402

Kursus ini merangkum jenis dan aljabar matriks, nilai eigen dan vektor eigen, taburan normal multivariat, serta min, varians dan taburan bentuk kuadratik. Penganggaraan dan pengujian hipotesis ke atas parameter model regresi linear pangkat penuh dan pangkat tak penuh, dan analisis korelasi menggunakan pendekatan matriks juga dibincangkan.

This course covers types and algebra of matrices, eigen values and eigen vectors, the multivariate normal distribution, as well as the mean and variance of distribution of quadratic forms. Estimation and hypothesis testing on parameters of full rank and non full rank linear regression models, and correlation analysis using matrix approach are also discussed.

MTH3405 Penggunaan Pakej Statistik Terpilih/ *Applications of Selected Statistical Package* 3(2+1)

Prasyarat : MTH3402

Kursus ini merangkum aplikasi pakej statistik terpilih untuk melaksanakan analisis pemerihalan, penganggaran dan inferens ke atas data.

This course covers applications of selected statistical package for performing descriptive analysis, estimation and inference on data.

MTH3406 Kawalan Kualiti Berstatistik/ *Statistical Quality Control* 3(3+0)

Prasyarat : MTH3402

Kursus ini merangkum teknik peningkatan kualiti dengan menggunakan statistik kawalan proses. Punca variasi, carta kawalan piawai Schewart, prosedur Cusum dan carta EWMA, analisis kebolehpayaan proses dan sistem pengukuran, ujikaji faktorasi bagi rekabentuk proses dan peningkatannya serta pensampelan penerimaan dibincangkan.

This course covers techniques for quality improvement through the use of statistical process control. Sources of variations, the standard Schewarts control chart, Cusum procedures and EWMA charts, process and measurement system capability analysis, factorial experiments for process design and improvement together with the acceptance sampling are discussed.

MTH3407 Kebarangkalian Pertengahan/ *Intermediate Probability* 3(3+0)

Prasyarat : MTH3402

Kursus ini merangkum kebarangkalian pada peringkat pertengahan. Tajuk yang dibincangkan termasuk sorotan kebarangkalian asas, persyaratan, ketaksamaan, fungsi ciri dan statistik tertib. Penumpuan dan teorem-teorem berkaitan dibincang.

This course covers probability at the intermediate level. Topics discussed include review of basic probability, conditioning, inequalities, characteristic function and order statistics. Convergence and the related theorems are discussed.

MTH3408 Pengenalan kepada Kaedah Bayes/ *Introduction to Bayesian Method* 3(3+0)

Prasyarat : MTH3402

Kursus ini merangkum konsep dan teori Bayes, taburan prior dan posterior, keluarga konjugat dan prior tak wajar, pentakbiran Bayes, peramalan, penganggaran titik dan selang berkredibiliti.

This course covers concepts and Bayesian theories, prior and posterior distribution, conjugate family and improper prior, Bayesian inference, predictions, point estimation and credibility interval.

MTH3409 Statistik Berkomputasi/ *Computational Statistics* 3(2+1)

Prasyarat : MTH3405

Kursus ini merangkum teori dan aplikasi teknik pengkomputeran dalam menyelesaikan masalah berstatistik dan melaksanakan simulasi berstatistik menggunakan bahasa pengaturcaraan terpilih.

This course covers the theory and application of computing techniques in solving statistical problems and performing statistical simulations using selected programming language.

MTH3410 Permodelan Dan Pentakbiran Berstatistik/ *Statistical Modelling and Inference* 3(3+0)

Prasyarat : MTH3402

Kursus ini memperkenalkan konsep dan pendekatan yang diperlukan untuk pemodelan berstatistik dan pemerihalan. Tajuk meliputi model berstatistik, fungsi kebolehjadian dan parameter

pentakbiran. Kursus ini juga membincangkan isu kenalaran dalam model, teori asimptot dan isu penganggar.

This course introduces the concepts and tools required for statistical modeling and inference. Topics include statistical models, likelihood functions and inference parameters. This course also discusses regularity in a model, asymptotic theory and estimator issue.

MTH3411 Analisis Regresi/ *Regression Analysis* 3(3+0)

Prasyarat : MTH3402

Kursus ini merangkum pembinaan model regresi linear mudah dan regresi linear berganda. Topik yang dibincangkan termasuk penganggaran parameter, ujian hipotesis, analisis varians, selang keyakinan, korelasi, analisis reja dan ramalan. Turut dibincangkan adalah model regresi polinomial dengan pembolehubah kualitatif.

This course covers simple and multiple linear regression model building. Topics discussed include parameter estimation, hypothesis testing, analysis of variance, confidence interval, correlation, residual analysis and prediction. Polynomial regression model with qualitative variable is also discussed.

MTH3500 Pengaturcaraan Komputer dalam Matematik/ *Computer Programming in Mathematics* 4(3+1)

Prasyarat : Tiada

Kursus ini merangkum pengaturcaraan komputer bagi menyelesaikan masalah matematik. Kemahiran membentuk algoritma, merekabentuk, mengkod, mengawal silap dan mendokumen aturcara dengan menggunakan teknik dan gaya pengaturcaraan yang betul serta berkesan diberi penekanan. Kaedah membina algoritma dan mereka bentuk aturcara dengan carta alir atau pseudo-kod dan konsep pengaturcaraan berstruktur dibincangkan.

This course covers computer programming for solving mathematical problems. Algorithm building skills, designing, coding, debugging and documenting using good and efficient programming techniques and styles are emphasized. Programme designing using flowcharts or pseudo-code and structured programming concept are discussed.

MTH3501 Analisis Berangka/ *Numerical Analysis* 3(3+0)

Prasyarat : MTH3500, MTH3102 dan MTH3201

Kursus ini merangkum kaedah interpolasi, penyelesaian berangka persamaan linear dan tak linear, penyelesaian berangka persamaan pembezaan biasa, pembezaan dan kamiran berangka dan analisis ralat.

This course covers methods of interpolation, numerical solution of linear and non-linear equations, numerical solution of ordinary differential equations, numerical differentiation and integration and error analysis.

MTH3602 Pengaturcaraan Bermatematik/ *Mathematical Programming* 3(3+0)

Prasyarat : MTH3102 dan MTH3201

Kursus ini merangkum teknik bermatematik yang digunakan sebagai alat untuk menyelesaikan masalah pemaksimuman atau meminimuman.

This course covers some mathematical techniques which are used as the tools for solving maximization or minimization problems.

MTH3701 Matematik Kewangan/ *Financial Mathematics* 3(3+0)

Prasyarat : MTH3100

Kursus ini merangkumi teori asas faedah. Penilaian faedah kuantitatif dianalisa merangkumi prinsip

asas yang terlibat dalam penilaian faedah. Prinsip asas ini diperluaskan kepada anuiti, pelunasan pinjaman, dana terikat dan penilaian bon.

This course covers the basic theory of interest. Quantitative measures of interest are analyzed which includes the basis principles involved in the measurement of interest. These basic principles are extended to annuities, loan amortization, sinking funds and valuation of bonds.

MTH3901	Proses Penyelidikan dalam Matematik dan Statistik/ <i>Research Processes in Mathematics and Statistics</i>	3(1+2)
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Prasyarat : MTH3500 atau MTH3405

Kursus ini merangkum proses dan kaedah penyelidikan serta kemahiran mencari maklumat. Teknik kreatif penyelesaian masalah dan pengenalan kepada kaedah matematik dan statistik dibincangkan. Kaedah penulisan saintifik, pembentangan dan penerbitan diterangkan.

This course covers research process and method and also information retrieval skills. The techniques of creative problem solving and introduction to mathematical and statistical methods are discussed. The methods of scientific writing, presentations and publications are described.

MTH4102	Teori Persamaan Pembezaan Biasa/ <i>Theory of Ordinary Differential Equations</i>	3(3+0)
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Prasyarat : MTH3102 dan MTH3301

Kursus ini merangkum teori kuantitatif tulen, teori kuantitatif hampiran dan teori kualitatif, teori kewujudan dan keunikan penyelesaian persamaan pembezaan biasa, dan teori Sturm-Liouville. Teori persamaan pembezaan matriks dan konsep matriks asasi dibangunkan. Konsep kestabilan dalam satah dan telatah penyelesaian persamaan pembezaan menggunakan kaedah langsung Liapunov turut dibincangkan.

This course covers pure quantitative theory, approximate quantitative theory and qualitative theory, theory of existence and uniqueness of solutions of ordinary differential equations and Sturm-Liouville theory. The theory of matrix differential equation and concept of fundamental matrices are developed. The concept of stability in the plane and the behaviour of the solutions of the differential equations are discussed by using Liapunov's direct method.

MTH4105	Teori Persamaan Kamiran/ <i>Theory of Integral Equations</i>	3(3+0)
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Prasyarat : MTH3102 dan MTH3301

Kursus ini merangkum persamaan kamiran linear dengan perbincangan ringkas mengenai persamaan kamiran tak linear. Tajuk yang dibincangkan termasuk klasifikasi persamaan kamiran, pertaliannya dengan persamaan pembezaan yang merangkum masalah nilai awal dan masalah nilai sempadan. Penyelesaian menggunakan kaedah penghampiran berturutan dan persamaan leraian, teori Fredholm dan teorem Hilbert-Schmidt dibincangkan.

The course covers linear integral equations with a brief discussion on simple non-linear equations. Topics discussed include the classification of integral equations, connection with differential equations which consist of initial value problems and boundary value problems. Solution by method of successive approximations and resolvent equations, Fredholm theory and Hilbert-Schmidt theorem are discussed.

MTH4106	Persamaan Pembezaan Separa/ <i>Partial Differential Equations</i>	3(3+0)
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Prasyarat : MTH3104 dan MTH3301

Kursus ini merangkum teori persamaan pembezaan separa dan kaedah penyelesaian. Persamaan pembezaan separa peringkat satu dan dua, serta bagaimana persamaan pembezaan separa digunakan dalam permasalahan fizik dibincangkan.

The course covers the theory of partial differential equations and methods for solution. First order

and second order partial differential equations, how the partial differential equations are used in physical problems are discussed.

MTH4201 Aljabar Niskala/ *Abstract Algebra*

3(3+0)

Prasyarat : MTH3202

Kursus ini merangkum konsep set, fungsi, set integer dan kekongruenan, kumpulan dan gelanggang, termasuk teori yang berkaitan dan diikuti dengan hasildarab terus kumpulan. Teori unggunan, operasi atas unggunan, beberapa jenis gelanggang, medan dan lanjutannya turut dibincangkan. Kursus ini berakhir dengan pengenalan unsur geometri algebra.

This course covers concepts related to set, functions, the set of integers and congruences, groups and rings including the related theories and followed by direct product of groups. Theory of ideals, operation on ideals, several types of rings, fields and their extensions are discussed. The course ends with introduction of elements of algebraic geometry.

MTH4202 Teori Nombor/ *Number Theory*

3(3+0)

Prasyarat : MTH3101 dan MTH3202

Kursus ini merangkum kebolehbahagian nombor integer, nombor perdana, takrifan dan kegunaan pembahagi terbesar sepunya, kekongruenan dan kesalingan kuadrat. Ini diikuti dengan penyelesaian persamaan Diofantus. Penggunaan Teori nombor bidang kriptografi turut dibincangkan.

This course covers the divisibility of integers, primes, definition and applications of greatest common divisor, congruence and quadratic reciprocity. This is followed by the solution of Diophantine equations. The applications of number theory in cryptography are discussed.

MTH4203 Teori Graf/ *Graph Theory*

3(3+0)

Prasyarat : MTH3202

Kursus ini merangkum graf Euleran dan Hamiltonan serta penggunaannya. Ini diikuti dengan pokok, kesatahan dan kedualan graf, nombor kromatik, mewarna peta dan pinggir, digraf, Teorem Hall, Teorem Menger dan kegunaannya.

This course covers Eulerian and Hamiltonian graphs and their applications. This is followed by trees, planar and dual graphs, chromatic number, map and edge colouring, diagraphs, Hall's theorem, Menger's theorem and their applications.

MTH4204 Kombinatorik/ *Combinatorics*

3(3+0)

Prasyarat : MTH3202

Kursus ini merangkum pengangkaan termasuk pilihatur dan gabungan, prinsip rangkuman dan eksklusi, persamaan linear berpekali unit, hubungan jadi semula dan fungsi penjana. Ini diikuti dengan kewujudan termasuk kaedah pembuktian, geometri satah, peta di atas sfera, masalah pewarnaan dan struktur terhingga. Kebarangkalian, percabangan teorem binomial, beberapa fungsi penjana dan persamaan beza, jujukan dan susunan Fibonacci turut dibincangkan.

This course covers enumeration including permutations and combinations, inclusion and exclusion principles, linear equations with unit coefficients, recursive relations and generating functions. This is followed by existence including methods of proofs, plane geometry, map on a sphere, colouring problems and finite structures. Probabilities, ramifications of binomial theorem, some generating functions and difference equations, Fibonacci sequences and arrangements are also discussed.

MTH4205 Kriptografi Bermatematik/ *Mathematical Cryptography*

3(3+0)

Prasyarat : MTH3202

Kursus ini merangkum konsep teori nombor, aljabar abstrak, gelanggang terhingga, teori informasi, teori kompleksiti dan teori kebarangkalian yang digunakan untuk memahami idea masalah log diskrit, kekuatan algoritma, keselamatan maklumat, enkripsi, dekripsi, sistem simetrik, asimetrik, tandatangan digital dan analisis kriptografi dalam kriptografi. Teori matematik kriptografi disebalik sistem krypto asimetrik, tandatangan digital dan simetrik diberi penekanan.

This course covers the concepts of number theory, abstract algebra, finite fields, information theory, complexity theory and probability theory to understand the ideas regarding the discrete log problem, strength of an algorithm, information security, encryption, decryption, symmetric systems, asymmetric systems, digital signatures and cryptanalysis in cryptography. The mathematical cryptographic theory behind asymmetric, digital signature and symmetric cryptosystems are emphasized.

MTH4301 Topologi/ *Topology*

3(3+0)

Prasyarat : MTH3301

Kursus ini merangkum konsep ruang, fungsi selanjar, set, kekardinalan set dan jenis set termasuk set terbuka dan tertutup. Ini diikuti dengan jujukan dalam ruang, topologi lemah dan kukuh, kekaitan, beberapa aksiom kekaitan dan jenis topologi. Teorem Lindeloff, Tychonoff dan Baire, ruang fungsi, ruang metrik dan ruang Baire dibincangkan.

This course covers concept of space, continuous function, set, cardinality of sets and types of sets which include open and closed sets. This is followed by sequences in space, weak and strong topologies, connectedness, axioms of connectedness, and types of topologies. The Lindeloff, Tychonoff and Baire theorems, function, metric, and Baire spaces are discussed.

MTH4302 Analisis Fungsian/ *Functional Analysis*

3(3+0)

Prasyarat : MTH3201 dan MTH3301

Kursus ini merangkum ruang metrik, ruang linear bernorma, ruang metrik padat, fungsian linear terbatas, ruang Hilbert, pengoperasi linear terbatas di atas ruang Hilbert dan teori spektrum bagi pengoperasi adjoint padat.

This course covers metric space, normed linear space, compact metric space, bounded linear functional, Hilbert space, bounded linear operators on Hilbert space and spectrum theory for compact adjoint operators.

MTH4401 Teknik Pensampelan/ *Sampling Techniques*

3(3+0)

Prasyarat : MTH3403

Kursus ini merangkum pelbagai teknik pensampelan yang sering digunakan dan aplikasinya. Pensampelan rawak mudah, pensampelan berstrata, pensampelan bersistematik, pensampelan berkelompok, anggaran nisbah, anggaran regresi dan kepincangan dalam pensampelan dibincangkan.

This course covers the various sampling techniques commonly used and their applications. Simple random sampling, stratified sampling, systematic sampling, cluster sampling, ratio and regression estimations and bias in sampling are discussed.

MTH4402 Tajuk Khas Dalam Statistik/ *Special Topics In Statistic*

(3+0)

Prasyarat : MTH3405

Tajuk terpilih dalam statistik ditentukan oleh Jabatan akan dibincangkan dengan lebih mendalam.

Selected topics in statistics determined by the Department will be discussed in depth in the lecture.

MTH4403 Statistik Tak Berparameter/ *Nonparametric Statistics* (3+0)

Prasyarat : MTH3403 dan MTH3404

Kursus ini merangkum kaedah tak berparameter dengan andaian kenormalan populasi dari mana sampel yang diperolehi, tidak dipenuhi. Kursus ini bermula dengan membincang secara ringkas kaedah tak berparameter serta perbandingannya dengan kaedah berparameter. Ini diikuti dengan perbincangan mengenai statistik tertib dan taburan kebarangkaliannya (tercantum dan sut), ujian rawakan, masalah lokasi dan serakan bagi sampel merdeka dan berkait, masalah kebagusan penyuaian dan sukatan sekutuan. Kursus ini di akhiri dengan membincangkan regresi linear mudah.

This course covers nonparametric methods in which the normality assumption of the population from which the samples are drawn is not met. The course begins with a brief discussion on the nonparametric methods and comparing them with the parametric methods. This is followed by the discussion on order statistics and their distributions (joint and marginal), test for randomness, location and dispersion problems for independent and related samples, problem on the goodness of fit and measure of association. The course ends with discussion on simple linear regression.

MTH4404 Proses Stokastik/ *Stochastic Processes* 3(3+0)

Prasyarat : MTH3402

Kursus ini merangkum beberapa jenis proses diskrit dan selanjar dengan penekanan diberi kepada rantai Markov dan perjalanan rawak.

This course covers several types of discrete and continuous processes with emphasis given to Markov chains and random walks.

MTH4405 Pengenalan kepada Analisis Multivariat/ *Introduction to Multivariate Analysis* 3(3+0)

Prasyarat : MTH3405

Kursus ini merangkum sifat asas bagi vektor rawak, teori taburan normal, penganggaran dan ujian hipotesis. Topik yang melibatkan analisis beberapa masalah multivariat dibincang.

This course covers the basic properties of random vectors, normal distribution theory, estimation and test of hypothesis. Topics involving analysis of several multivariate problems are discussed.

MTH4406 Siri Masa/ *Time Series* 3(3+0)

Prasyarat : MTH3404

Kursus ini merangkum penganggaran trend dan pola bermusim, proses pegun dan proses ARMA. Pengenalpastian, anggaran, diagnosis dan ujian rawakan, kriteria pemilihan, peramalan siri masa pegun, beberapa algoritma dan model pola bermusim bersifat darab dibincangkan.

This course covers estimation of trend and seasonal patterns, stationary and ARMA processes. Identification, estimation, diagnostic and randomness test, order forecasting stationary time series, several algorithms and multiplicative seasonal models are discussed.

MTH4407 Kaedah Interaktif Berkomputasi dalam Analisis Data/ *Interactive Computational Methods In Data Analysis* 3(3+0)

Prasyarat : MTH3405

Kursus ini merangkum penganalisan dan pentafsiran hasil daripada pakej berstatistik terpilih. Transformasi dalam model berstatistik, pengujian andaian, analisis reja, data terpencil, simulasi berstatistik dan kegunaan varians serta kajian kes secara berkumpulan ke atas contoh yang berkaitan dibincangkan.

The course covers analyses and interpretations of output from selected statistical package. Transformations in statistical models, assumptions checking, residual analysis, outliers, statistical simulations and applications of variance as well as group case studies on related examples are discussed.

MTH4408 Pengenalan kepada Analisis Mandirian/*Introduction to Survival Analysis* 3(3+0)

Prasyarat : MTH3405

Kursus ini memperkenalkan konsep dan pendekatan yang diperlukan untuk menganalisis data mandirian. Tajuk meliputi teknik jelajah, berparameter dan berparameter separa. Kursus ini juga melibatkan penggunaan pakej statistik yang bersesuaian dalam analisis.

This course introduces the concepts and tools required for the analysis of survival data. The topics include exploratory, parametric and semi-parametric techniques. This course also involves the usage of appropriate statistical package in the analysis.

MTH4501 Analisis Berangka Lanjutan/*Advanced Numerical Analysis* 3(3+0)

Prasyarat : MTH3501

Kursus ini meliputi teknik menyelesaikan masalah persamaan pembezaan secara berangka menggunakan kaedah satu langkah dan multi langkah. Kestabilan sifar dan kestabilan mutlak turut dikaji. Kaedah berangka bagi menyelesaikan persamaan pembezaan separa turut dibincangkan.

This course covers the techniques for solving differential equations numerically using one-step and multistep methods. Zero and absolute stability will be investigated. Numerical methods for solving partial differential equations are also discussed.

MTH4502 Teori Penghampiran/*Approximation Theory* 3(3+0)

Prasyarat : MTH3602

Kursus ini merangkum kewujudan dan keunikan suatu penghampiran umum, dan penghampiran terbaik dalam norma seragam. Ini diikuti dengan pembentukan penghampiran menggunakan polinomial ortogon dan penghampiran menggunakan fungsi nisbah.

This course covers the existence and uniqueness of approximations, and the best approximation in the uniform norm. This is followed by the constructions of the approximations using orthogonal polynomials and the approximation using rational functions.

MTH4602 Kawalan Optimum/*Optimal Control* 3(3+0)

Prasyarat : MTH3104

Kursus ini merangkum analisis dan rekabentuk sistem dinamik yang rumit. Teori kawalan optimum, pengaturcaraan dinamik, prinsip Pontryagin dan sistem kawalan linear dibincangkan.

This course covers the analysis and design of complicated dynamic systems. The optimal control theory, dynamic programming, Pontryagin's principles and linear control systems are discussed.

MTH4603 Penyelidikan Operasi/*Operations Research* 3(3+0)

Prasyarat : MTH3602

Kursus ini merangkum analisis, teknik dan pemodelan bermatematik dalam bidang penyelidikan operasi. Masalah pengangkutan, model rangkaian, model inventori, dan sistem giliran dibincangkan.

This course covers analysis, technique and mathematical modeling in the field of operations research. Transportation problems, network models, inventory models, and queuing systems are discussed.

MTH4604 Teknik Pengoptimuman/ Optimization Techniques 3(3+0)

Prasyarat : MTH3401 dan MTH3201

Kursus ini merangkum teori permulaan yang menjadi asas kepada teknik pengoptimuman terkini. Konsep teori yang mendalam dan penggunaan sebenar teknik pengoptimuman ditekankan.

This course covers an elementary theory on which the current optimization techniques are based. The detailed theoretical concepts and the actual application of optimization techniques are emphasized.

MTH4605 Teori Kawalan/ Control Theory 3(3+0)

Prasyarat : MTH3104 dan MTH3301

Kursus ini merangkum pendekatan baharu dan terkini teori kawalan linear klasik, pengetahuan asas bagi analisis dan rekabentuk otomatik atau gelung tertutup sistem kawalan.

This course covers a new and current approaches on classical linear control theory, basic knowledge of analysis and automatic design, or closed loop of control systems.

MTH4606 Tajuk Khas dalam Matematik Gunaan/ Special Topics In Applied Mathematics 3(3+0)

Prasyarat : MTH3104

Kursus ini membincangkan tajuk terkini dalam matematik gunaan.

This course discusses the current topics in applied mathematics.

MTH4800 Sejarah Matematik/ History of Mathematics (3+0)

Prasyarat : MTH3301

Kursus ini merangkum pembentukan idea matematik yang melibatkan teori matematik moden. Aspek kualitatif dan kuantitatif berdasarkan perspektif sejarah, pembentukan sejarah dalam cabang penting matematik termasuklah teori nombor, aljabar, geometri dan logik dibincangkan.

This course covers the development of mathematical ideas which is related to the theory of modern mathematics. Both qualitative and quantitative aspects based on historical perspective, historical development in some important branches of mathematics including number theory, algebra, geometry and logic are discussed.

MTH4903 Latihan Industri/ Industrial Training 8(0+8)

Prasyarat : MTH4959

Kursus ini meliputi latihan industri selama 16 minggu di sektor awam/swasta bagi mempraktikkan ilmu yang diperolehi dalam program pengajian.

This course covers industrial training for a period of 16 weeks at government/private sectors to apply the knowledge acquired in the programme of study.

MTH4959 Disertasi Bachelo/ Bachelor Dissertation 6(0+6)

Prasyarat : MTH3901

Kursus ini merangkum sorotan kepustakaan, metodologi penyelidikan yang sesuai, pengumpulan dan analisis data, pentafsiran keputusan, perbincangan dan kesimpulan kajian saintifik serta pembentangan hasil penyelidikan.

This course covers literature review, appropriate research methodology, data collection and analysis, interpretation of results, discussion and conclusion of scientific studies and presentation of research output.

(ii) Bacelor Sains Matematik dengan Kepujian & Bacelor Sains Statistik dengan Kepujian / Bachelor of Science in Mathematics with Honours & Bachelor of Science in Statistics with Honours

MTS3101 Kalkulus I/ Calculus I

3(3+0)

Prasyarat : Tiada

Kursus ini merangkum pembinaan konsep dalam kalkulus satu pemboleh ubah, konsep set dan fungsi untuk memahami idea mengenai had, kesinambungan dan terbitan. Pembezaan dan teorem yang berkaitan kamiran sebagai proses anti-terbitan serta teknik kamiran ditekankan.

This course covers the building up of the concepts in calculus of one variable, the concept of sets and functions to understand the idea of limits, continuity and derivatives. Differentiations and theorems related to integration as a process of anti-derivatives together with the integration techniques are emphasized.

MTS3102 Kalkulus II/ Calculus II

3(3+0)

Prasyarat : MTS3101

Kursus ini merangkum pelbagai siri dan jujukan serta fungsi, persamaan parametrik, koordinat kutub dan geometri vektor. Kalkulus bagi fungsi nilai vektor, fungsi banyak pemboleh ubah yang melibatkan kalkulus pembezaan dalam beberapa pemboleh ubah dibincangkan.

This course covers various series and sequences as well as functions, parametric equations, polar coordinates and vector geometry. Calculus for vector value functions, functions of many variables involving differential calculus in several variables are discussed.

MTS3103 Kalkulus III/ Calculus III

3(3+0)

Prasyarat : MTS3102

Kursus ini merangkum konsep fungsi multi pemboleh ubah dan fungsi vektor. Kaedah kamiran berganda dan teorem asas kalkulus vektor serta penggunaannya turut dibincangkan.

This course covers the concepts of functions in multivariables and vector. Method of multiple integration and fundamental theorem of calculus together with its application are also discussed.

MTS3104 Persamaan Pembezaan/ Differential Equations

3(3+0)

Prasyarat : MTS3101 atau MTS3100

Kursus ini merangkum pengelasan persamaan pembezaan, kaedah penyelesaian persamaan pembezaan linear dan kaedah membina penyelesaian am daripada beberapa penyelesaian khusus yang diperolehi. Persamaan pembezaan tak homogen diselesaikan menggunakan pelbagai kaedah serta kaedah penyelesaian bagi sistem persamaan pembezaan dibincangkan.

This course covers the classification of differential equations, methods of solving linear differential equations and methods of constructing general solutions from several specific solutions obtained. Nonhomogeneous differential equations are solved using various methods as well as solution methods for the system of differential equations are discussed.

MTS3201 Aljabar/ Algebra

3(3+0)

Prasyarat : Tiada

Kursus ini merangkum konsep dan kaedah penulisan bermatematik dalam aljabar. Penyelesaian masalah berkaitan dengan polinomial, sistem nombor nyata dan nombor kompleks dengan kaedah yang sesuai dibincangkan.

This course covers the concepts and methods of mathematical writing in algebra. Problem solving related to polynomials, real number systems and complex numbers with appropriate methods is discussed.

MTS3202 Aljabar Linear/ *Linear Algebra* 3(3+0)

Prasyarat : MTS3101 dan MTS3201

Kursus ini merangkum konsep asas aljabar linear, masalah nilai eigen dan vektor eigen. Transformasi linear, keserupaan dan pepenjuru bagi suatu matrik dibincangkan. Penyelesaian masalah matematik dengan konsep aljabar linear dibincangkan.

This course covers the basic concepts of linear algebra, eigenvalue problems and eigenvectors. Linear transformations, similarity and diagonality of a matrix are discussed. Mathematical problems solving using the concept of linear algebra is discussed.

MTS3203 Aljabar Modern/ *Modern Algebra* 3(3+0)

Prasyarat : MTS3202

Kursus ini merangkum konsep kebolehbahagian dan pembahagi sepunya terbesar. Menganalisis masalah bagi kumpulan, gelanggang, medan dan domain integer turut dibincangkan.

This course covers the concepts of divisibility and the largest common divisor. Analysing problems of group, ring, field and integral domain are also discussed.

MTS3301 Analisis Nyata/ *Real Analysis* 3(3+0)

Prasyarat : MTS3102

Kursus ini merangkum konsep jujukan, ruang metrik dan kamiran. Analisis masalah yang berkaitan dengan ruang metrik pelengkap dan padat dibincangkan.

This course covers the concepts of sequence, metric space and integrals. Analysis of problems related to complement and compact metric spaces is discussed.

MTS3302 Analisis Kompleks/ *Complex Analysis* 3(3+0)

Prasyarat : MTS3102

Kursus ini merangkum teorem asas dalam analisis kompleks. Penyelesaian masalah bagi kamiran fungsi kompleks sepanjang pelbagai lintasan dalam satah kompleks dibincangkan.

This course covers basic theorems in complex analysis. The solution for the integral problem of a complex function along various path of integration in a complex plane is discussed.

MTS3500 Pengaturcaraan Komputer dalam Matematik/ *Computer Programming in Mathematics* 4(3+1)

Prasyarat : Tiada

Kursus ini merangkum bahasa pengaturcaraan komputer bagi menyelesaikan masalah bermatematik. Pembinaan algoritma dan aturcara untuk penyelesaian masalah berkaitan turut dibincangkan.

This course covers computer programming languages for solving mathematical problems. The construction of algorithms and programs for solving related problems are also discussed.

MTS3501 Analisis Berangka/ *Numerical Analysis* 3(3+0)

Prasyarat : MTS3500 dan MTS3104 dan MTS3202

Kursus ini merangkum pelbagai kaedah berangka dalam penyelesaian masalah matematik. Perbincangan termasuk ralat, penyelesaian persamaan linear dan tak linear, interpolasi, pembezaan dan pengamiran berangka, kaedah satu langkah dan multilangkah.

This course covers various numerical methods in solving mathematical problems. Discussion includes error, solution of linear and non-linear equations, interpolation, numerical differentiation and integration, one-step and multistep methods.

MTS3502 Pakej Perisian dan Penggunaan dalam Matematik/ *Software Packages And Application In Mathematics* 3(2+1)

Prasyarat : Tiada

Kursus ini merangkum kepelbagaian pakej perisian bagi menyelesaikan masalah bermatematik. Pembinaan algoritma dan aturcara serta penggunaannya untuk tujuan analisis dan pengiraan dibincangkan.

This course covers a variety of software packages for solving mathematical problems. The construction of algorithms and programs and their use for analysis and computational purposes are discussed.

MTS3602 Pengaturcaraan Bermatematik/ *Mathematical Programming* 3(3+0)

Prasyarat : MTS3104 dan MTS3202

Kursus ini merangkum masalah pengaturcaraan bermatematik dan model matematik yang terbit daripada masalah pengoptimuman. Pembinaan pengaturcaraan bermatematik bagi penyelesaian masalah optimum dibincangkan.

This course covers mathematical programming problems and mathematical models arising from optimization problems. The construction of mathematical programming for optimal problem solving is discussed.

MTS3701 Matematik Kewangan/ *Financial Mathematics* 3(3+0)

Prasyarat : MTS3100 atau MTS3101

Kursus ini merangkum konsep matematik kewangan dalam kehidupan seharian. Kemahiran keusahawanan dan perniagaan melalui penyelesaian masalah berkaitan matematik kewangan dibincang.

This course covers the mathematical concepts of finance in everyday life. Entrepreneurial and business skills through problem solving related to financial mathematics are discussed.

MTS3901 Proses Penyelidikan dalam Matematik/ *Research Processes in Mathematics* 3(2+1)

Prasyarat : MTS3500

Kursus ini merangkum proses dan kaedah penyelidikan serta kemahiran mencari maklumat, teknik kreatif penyelesaian masalah dan pengenalan kepada kaedah penyelesaian masalah matematik. Kaedah penulisan saintifik, pembentangan dan penerbitan dibincangkan.

This course covers research processes and methods as well as information seeking skills, creative problem-solving techniques and an introduction to mathematical solution methods. Methods of scientific writing, presentation and publication are discussed.

MTS4102 Teori Persamaan Pembezaan Biasa/ *Theory of Ordinary Differential Equations* 3(3+0)

Prasyarat : MTS3104

Kursus ini merangkum teorem kewujudan dan keunikan serta penyelesaian persamaan pembezaan biasa. Mengenal pasti kaedah telatah penyelesaian masalah nilai awal dan nilai sempadan serta aplikasinya dalam bidang berkaitan ditekankan.

This course covers the existence and uniqueness theorem and the characteristics of the solutions to the ordinary differential equations. Identifying the methods for solving the initial value and boundary problems and their applications in the relevant field are emphasized.

MTS4105 Teori Persamaan Kamiran/ *Theory of Integral Equations* 3(3+0)

Prasyarat : MTS3104 dan MTS3301

Kursus ini merangkum konsep dan teori persamaan kamiran. Penyelesaian menggunakan kaedah penghampiran berturutan dan persamaan leraian, teori Fredholm dan teorem Hilbert-Schmidt dibincangkan.

This course covers the concept and theory of integral equations. Solutions using sequential approximation methods and solution equations, Fredholm theory and Hilbert-Schmidt theorem are discussed.

MTS4106 Persamaan Pembezaan Separat/ *Partial Differential Equations* 3(3+0)

Prasyarat : MTS3104

Kursus ini merangkum konsep dan teori persamaan pembezaan separa dan kaedah penyelesaian. Penyelesaian persamaan pembezaan separa peringkat satu dan dua, serta pengaplikasiannya dalam permasalahan fizik ditekankan.

This course covers concepts and theories of partial differential equations and methods of solution. Solutions of first and second order partial differential equations, and its application in physics problems are emphasized.

MTS4107 Mekanik Bendalir/ *Fluid Mechanics* 3(3+0)

Prasyarat : MTS3103

Kursus ini merangkum teori persamaan mekanik bendalir dan konsep hukum keabadian dalam sistem bendalir. Pemerihalan gerakan bendalir serta analisis pembezaan dan kamiran gerakan bendalir dibincangkan.

The course covers the theory of equations of fluid mechanics and the concept of conservation law in fluid system. The description of fluid motion as well as the differential and integral analysis of fluid motion are discussed.

MTS4201 Aljabar Niskala/ *Abstract Algebra* 3(3+0)

Prasyarat : MTS3203

Kursus ini merangkum konsep asas dalam aljabar. Penekanan diberi kepada teori unggunan, operasi atas unggunan, beberapa jenis gelanggang, medan dan lanjutannya.

The course covers basic concept in algebra. Emphasis is also given to the theory of ideals, operation on ideals, several types of rings, fields and their extensions.

MTS4202 Teori Nombor/ *Number Theory* 3(3+0)

Prasyarat : MTS3203

Kursus ini merangkum konsep teori nombor, penggunaan konsep kekongruenan dan pembahagi terbesar sepunya. Permasalahan berkaitan kesalingan kuadrat dan penggunaan teori nombor dalam kriptografi dibincangkan.

This course covers the concepts of number theory, applications of the concept of congruence and greatest common divisors. Problems related to quadratic reciprocity and the applications of number theory in cryptography are discussed.

MTS4203 Teori Graf/ *Graph Theory* 3(3+0)

Prasyarat : MTS3203

Kursus ini merangkum graf Euleran dan Hamiltonan serta penggunaannya. Ini diikuti dengan pokok, kesatahan dan kedualan graf, nombor kromatik, mewarna peta dan pinggir, digraf, Teorem Hall, Teorem Menger dan kegunaannya.

This course covers Eulerian and Hamiltonian graphs and their applications. This is followed by trees, planar and dual graphs, chromatic number, map and edge colouring, digraphs, Hall's theorem, Menger's theorem and their applications.

MTS4204 Kombinatorik/ *Combinatorics* 3(3+0)

Prasyarat : MTS3203

Kursus ini merangkum konsep dan teorem dalam kombinatorik. Menyelesaikan masalah pengangkaan dan kewujudan serta aplikasinya dalam bidang berkaitan dibincangkan.

This course covers concepts and theorems in combinatorics. Solving numeracy problems and their existence also its application in related fields are discussed.

MTS4210 Matematik Kriptografi/ *Cryptography Mathematics* 3(3+0)

Prasyarat : MTS3203

Kursus ini merangkum konsep sistem kriptografi dan pembinaan algoritma bagi proses enkripsi dan dekripsi melalui suatu proses kriptografi bermatematik. Kajian berkaitan peluang keusahawanan berdasarkan kepentingan dan penggunaan kriptografi bermatematik dalam sistem keselamatan informasi dibincangkan.

This course covers the concept of cryptographic systems and the construction of algorithms for the process of encryption and decryption through a mathematical cryptographic process. Studies related to entrepreneurial opportunities based on the importance and use of mathematical cryptography in information security systems are discussed.

MTS4211 Sistemkripto RSA/ *RSA Cryptosystem* 3(3+0)

Prasyarat : MTS4210

Kursus ini merangkum konsep matematik dan statistik untuk memahami proses yang dilakukan oleh sistemkripto RSA dan variasinya serta aplikasi sistemkripto RSA sebagai sistem enkripsi dan tandatangan digital. Teknik analisis-kripto terhadap sistemkripto RSA dan idea di sebaliknya yang menggunakan teori matematik kriptografi turut diberi penekanan.

This course covers the concepts of mathematics and statistics to understand the processes undertaken by the RSA cryptosystem and its variants along with its application as encryption and digital signature systems. The cryptanalysis techniques against the RSA cryptosystem and the idea behind them involving mathematical cryptographic theory are emphasized.

MTS4212 Sistem Kripto Lengkung Eliptik/ *Elliptic Curve Cryptosystem* 3(3+0)

Prasyarat : MTS4210

Kursus ini merangkum sistem kripto lengkung Eliptik (ECC) dan kaedah serangan terhadap logaritma diskritnya yang berkonsepkan kriptografi bermatematik. Kaedah pendaraban skalar dan tiga jenis serangan terhadap logaritma diskrit lengkung Eliptik dibincangkan.

This course covers the Elliptical curve crypto system (ECC) and the method of attack on its discritical logarithm based on the concept of mathematical cryptography. Scalar multiplication methods and three types of attacks on Elliptic curve discritical logarithms are discussed.

MTS4213 Kriptografi Pasca Kuantum/ *Post Quantum Cryptography* 3(3+0)

Prasyarat : MTS4210

Kursus ini merangkum konsep kriptografi pasca-kuantum untuk memahami idea kekuatan dalam pembinaan algoritma penyulitan, penyahsulitan, dan tandatangan digital kriptosistem pasca-kuantum. Teori matematik kriptografi di sebalik pembinaan kriptosistem dan analisis-kripto Galas, GGH, dan NTRU dibincangkan.

This course covers the concept of post-quantum cryptography to understand the idea of strengths in the construction of encryption, decryption, and digital signature algorithms of post-quantum cryptosystems. The mathematical theory of cryptography behind cryptosystem construction and crypto-analysis Galas, GGH, and NTRU are discussed.

MTS4214 Teori Pengkodan/ *Coding Theory* 3(3+0)

Prasyarat : MTS4210

Kursus ini merangkum konsep matematik kod linear berserta medan terhingga dalam membincangkan sistem komunikasi yang berkaitan pengkodan, nyahkod dan pengendalian ralat secara holistik. Syarat-syarat parameter kod sedemikian ianya wujud, teori pengkodan Reed-Solomon berkenaan pengkodan dan nyahkod dan juga aplikasinya diberi penekanan dalam kursus ini.

This course discusses mathematical concepts of linear codes together with finite fields in order to holistically discuss communication systems related to coding, decoding and error handling. The conditions on the parameters of codes such that it exists, coding theory of Reed-Solomon encoding and decoding and its applications are emphasized in this course.

MTS4301 Topologi/ *Topology* 3(3+0)

Prasyarat : MTS3301

Kursus ini merangkum pelbagai bentuk ruang matrik, ruang topologi serta beberapa teorem utama yang berkaitan. Penyelesaian masalah berkaitan ruang matrik dan ruang topologi serta aplikasinya dibincangkan.

This course covers various forms of matrix spaces, topological spaces as well as some key related theorems. Problem solving related to matrix space and topology space as well as their applications are discussed.

MTS4302 Analisis Fungsian/ *Functional Analysis* 3(3+0)

Prasyarat : MTS3202 dan MTS3301

Kursus ini merangkum konsep dan teorem analisis fungsian. Teori Spektrum bagi pengoperasian adjoint padat dan geometri ruang Hilbert dibincangkan.

This course summarizes the concepts and theorems of functional analysis. Spectrum theory for dense adjoining operations and Hilbert space geometry are discussed.

MTS4501 Analisis Berangka Lanjutan/ *Advanced Numerical Analysis* 3(3+0)

Prasyarat : MTS3501

Kursus ini meliputi teknik menyelesaikan masalah persamaan pembezaan biasa secara berangka. Analisis kejutan dan ralat yang terhasil daripada penyelesaian masalah persamaan pembezaan secara berangka turut dibincangkan.

This course covers techniques for solving ordinary differential equation problems numerically. The analysis of accuracy and error resulting from the solution of numerical differential equation problems is also discussed.

- MTS4502 Teori Penghampiran/ *Approximation Theory* 3(3+0)
- Prasyarat : MTS3602
- Kursus ini merangkum kewujudan dan keunikan anggaran dan penghampiran terbaik dalam norma seragam. Pembinaan daripada penghampiran menggunakan polinomial ortogon dan penghampiran menggunakan fungsi rasional dibincang.*
- This course covers the existence and uniqueness of estimates and best approximations in uniform norms. Constructions of approximations using orthogonal polynomials and approximations using rational functions are discussed.*
- MTS4503 Perisian Bermatematik/ *Mathematical Software* 3(2+1)
- Prasyarat : Tiada
- Kursus ini merangkum penggunaan perisian matematik terpilih untuk menyelesaikan masalah sains matematik. Pengenalan kepada kemampuan pakej perisian matematik pengiraan interaktif secara simbolik dan berangka, analisis data dan grafik serta penggunaannya dibincangkan.*
- This course covers the use of selected mathematical software to solve the problem of mathematics science. Introduction to the ability of interactive mathematical software packages to calculate symbolically and numerically, data and graphic analysis as well as their usage are discussed.*
- MTS4602 Kawalan Optimum/ *Optimal Control* 3(3+0)
- Prasyarat : MTS3103
- Kursus ini merangkum kawalan optimum, telatah optimum, indeks prestasi dan sistem statik dan dinamik. Teori kawalan optimum, pengaturcaraan dinamik, prinsip Pontryagin dan sistem kawalan linear dibincangkan.*
- The course covers optimal control, optimal antics, performance indices and static and dynamic systems. Optimal control theory, dynamic programming, Pontryagin principles and linear control systems are discussed.*
- MTS4603 Penyelidikan Operasi/ *Operations Research* 3(3+0)
- Prasyarat : MTS3602
- Kursus ini merangkum analisis, teknik dan pemodelan bermatematik dalam bidang penyelidikan operasi. Masalah pengangkutan, model rangkaian, model inventori, dan sistem giliran dibincangkan.*
- This course covers analysis, technique and mathematical modeling in the field of operations research. Transportation problems, network models, inventory models, and queuing systems are discussed.*
- MTS4604 Teknik Pengoptimuman/ *Optimization Techniques* 3(3+0)
- Prasyarat : MTS3103 dan MTS3202
- Kursus ini merangkum teori permulaan yang menjadi asas kepada teknik pengoptimuman terkini. Konsep teori yang mendalam dan penggunaan sebenar teknik pengoptimuman ditekankan.*
- This course covers an elementary theory on which the current optimization techniques are based. The detailed theoretical concepts and the actual application of optimization techniques are emphasized.*
- MTS4605 Teori Kawalan/ *Control Theory* 3(3+0)
- Prasyarat : MTS3103 dan MTS3301
- Kursus ini merangkum konsep teori kawalan dan reka bentuk automatik atau gelung tertutup sistem kawalan. Pendekatan baharu dan terkini teori kawalan linear klasik dan penyelesaian masalah teori kawalan dibincangkan.*

This course covers the concepts of control theory and the design of automatic or closed loop control systems. New and recent approaches to classical linear control theory and problem solving of control theory are discussed.

MTS4800 Sejarah Matematik/ *History of Mathematics* 3(3+0)

Prasyarat : MTS3301

Kursus ini merangkum sumbangan tokoh yang terkenal dalam sejarah matematik. Aspek kualitatif dan kuantitatif berdasarkan perspektif sejarah, pembentukan sejarah dalam cabang penting matematik dibincangkan.

This course covers the contributions of well-known figures in the history of mathematics. Qualitative and quantitative aspects based on historical perspective, historical formation in important branches of mathematics are discussed.

MTS4903 Latihan Industri/ *Industrial Training* 8(0+8)

Prasyarat : Dengan kebenaran jabatan

Kursus ini merangkumi pendedahan kepada persekitaran kerja yang sebenar di industri/organisasi. Aplikasi teori, praktikal, kemahiran komunikasi dan penyelesaian masalah turut ditekankan.

This course covers the exposure to real working environment in industries/organizations. Application of the theoretical, practical, communication and problem solving skills are also emphasized.

MTS4959 Disertasi Bachelo/ *Bachelor Dissertation* 6(0+6)

Prasyarat : Dengan kebenaran jabatan

Kursus ini merangkum sorotan kepustakaan, metodologi penyelidikan yang sesuai, pengumpulan dan analisis data, pentafsiran keputusan, perbincangan dan kesimpulan kajian saintifik serta pembentangan hasil penyelidikan.

This course covers literature review, appropriate research methodology, data collection and analysis, interpretation of results, discussion and conclusion of scientific studies and presentation of research output.

STS3401 Kebarangkalian dan Statistik I/ *Probability and Statistics I* 3(3+0)

Prasyarat : MTS3101

Kursus ini merangkum teori kebarangkalian dan kaedah statistik yang bersesuaian. Penyelesaian masalah kebarangkalian dan statistik menggunakan kaedah yang sesuai turut dibincangkan.

This course covers probability theory and appropriate statistical methods. Probability and statistical problem solving using appropriate methods are also discussed.

STS3402 Kebarangkalian dan Statistik II/ *Probability and Statistics II* 3(3+0)

Prasyarat : STS3401

Kursus ini merangkum konsep kebarangkalian dan pelbagai taburan kebarangkalian termasuk penerbitan dan aplikasi beberapa fungsi penjana. Penekanan juga diberi kepada penjelmaan pemboleh ubah dan penganggaran.

This course covers probability concept and various probability distributions including generating functions and their applications. Emphasis is also given to variable transformation and estimation.

STS3403 Rekabentuk Ujikaji/ *Experimental Design* 3(3+0)

Prasyarat : STS3401

Kursus ini merangkum teknik analisis varians bagi beberapa rekabentuk yang penting. Pembahagian jumlah kuasa dua, teknik perawakan, menerbitkan statistik ujian dan kaedah

perbandingan berganda ditekankan.

This course covers analysis of variance techniques for several important designs. The partitioning of the total sum of squares, randomization techniques, derivation of test statistics and multiple comparison methods are emphasized.

STS3404 Sains Data dengan Aplikasi/ *Data Science with Applications* 3(2+1)

Prasyarat : Tiada

Kursus ini membuat penerokaan dan pelbagai analisis sains data dari kumpulan data yang besar. Pengetahuan teori dan kemahiran yang dipelajari akan diaplikasikan dalam data sebenar.

This course makes exploration and various data science analysis of large data sets. Theoretical knowledge and skills learned will be applied in real data.

STS3405 Penggunaan Pakej Statistik Terpilih/ *Applications of Selected Statistical Package* 3(2+1)

Prasyarat : STS3401

Kursus ini merangkum prosedur dalam menggunakan perisian statistik terkini dalam menyelesaikan masalah statistik. Analisis pemerihalan, penganggaran dan pentaabiran ke atas data turut dibincangkan.

This course covers procedures of using latest statistical software in solving statistical problems. Descriptive analysis, estimation and estimation of the data are also discussed.

STS3406 Kawalan Kualiti Berstatistik/ *Statistical Quality Control* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum teknik peningkatan kualiti dengan menggunakan kawalan kualiti berstatistik. Masalah berkaitan jenis carta kawalan dan analisis kebolehpayaan proses yang sesuai digunakan turut dibincangkan.

This course covers quality improvement techniques using statistical quality control. Problems related to the type of control chart and the appropriate process capability analysis to be used are also discussed.

STS3407 Kebarangkalian Pertengahan/ *Intermediate Probability* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum kebarangkalian pada peringkat pertengahan. Teorem berkaitan dengan kebarangkalian bermatematik turut dibincangkan.

This course covers probability at the intermediate level. Theorems related to mathematical probabilities are also discussed.

STS3408 Kaedah Bayes/ *Bayesian Method* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum konsep konsep Bayes sebagai satu bidang penyelidikan dalam statistik. Penghasilan semula penyelesaian masalah menggunakan kaedah Bayes turut dibincangkan.

This course covers the concept of Bayes as an area of research in statistics. The reproduction of problem solving using the Bayes method is also discussed.

STS3409 Statistik Berkomputasi/ *Computational Statistics* 3(2+1)

Prasyarat : STS3402 dan STS3405

Kursus ini merangkum teori dan aplikasi teknik pengkomputeran dalam menyelesaikan masalah berstatistik dan melaksanakan simulasi berstatistik menggunakan bahasa pengaturcaraan terpilih.

This course covers the theory and application of computational techniques in solving statistical problems and performing statistical simulations using selected programming language.

STS3410 Permodelan Dan Pentaabiran Berstatistik/ *Statistical Modelling and Inference* 3(3+0)

Prasyarat : STS3402

Kursus ini memperkenalkan konsep dan pendekatan yang diperlukan untuk pemodelan berstatistik dan pemerihalan. Kaedah tertentu dan alternatif bagi menyelesaikan masalah berstatistik dalam pemodelan dan pentaabiran statistik turut dibincangkan.

This course introduces the concepts and tools required for statistical modeling and inference. Specific and alternative methods for solving statistical problems in statistical modeling and interpretation are also discussed.

STS3411 Analisis Regresi/ *Regression Analysis* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum pembinaan model regresi linear mudah dan regresi linear berganda. Penganggaran parameter bagi model regresi yang bersesuaian untuk menganalisis data dan menyelesaikan masalah yang berkaitan turut dibincangkan.

This course covers simple and multiple linear regression model building. Parameter estimation for appropriate regression models for analyzing data and solving related problems is also discussed.

STS3412 Analisis Data Multivariat/ *Multivariate Data Analysis* 3(3+0)

Prasyarat : STS3409

Kursus ini merangkum sifat asas vektor rawak, teori taburan normal, penganggaran dan ujian hipotesis. Analisis beberapa masalah multivariat dibincangkan.

This course covers the basic properties of random vectors, normal distribution theory, estimation and test of hypothesis. Analysis of several multivariate problems are discussed.

STS3901 Proses Penyelidikan dalam Statistik/ *Research Processes in Statistics* 3(2+1)

Prasyarat : STS3405

Kursus ini merangkum proses dan kaedah penyelidikan serta kemahiran mencari maklumat. Teknik kreatif penyelesaian masalah dan pengenalan kepada kaedah statistik dibincangkan. Kaedah penulisan saintifik, pembentangan dan penerbitan diterangkan.

This course covers research process and methods, and information retrieval skill. The techniques of creative problem solving and introduction to statistical methods are discussed. The methods of scientific writing, presentations and publications are described.

STS4401 Teknik Pensampelan/ *Sampling Techniques* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum pelbagai teknik pensampelan dan aplikasinya. Kaedah pensampelan dan pentaabiran yang berkaitan turut dibincangkan.

This course covers various sampling techniques and their applications. Relevant sampling and estimation methods are also discussed.

STS4402 Tajuk Khas Dalam Statistik/ *Special Topics In Statistic* 3(3+0)

Prasyarat : STS3405

Tajuk terpilih dalam statistik ditentukan oleh Jabatan akan dibincangkan dengan lebih mendalam.

Selected topics in statistics determined by the Department will be discussed in depth in the lecture.

STS4403 Statistik Tak Berparameter/ *Nonparametric Statistics* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum kaedah statistik tak berparameter bagi masalah lokasi dan serakan. Analisis data menggunakan kaedah statistik tak berparameter turut dibincangkan.

This course covers nonparametric statistical methods for location and dispersion problems. Data analysis using nonparametric statistical methods for are also discussed.

STS4404 Proses Stokastik/ *Stochastic Processes* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum beberapa jenis proses diskrit dan selanjut dengan penekanan diberi kepada rantai Markov dan perjalanan rawak.

This course covers several types of discrete and continuous processes with emphasis given to Markov chains and random walks.

STS4405 Analisis Data Berkategori Gunaan/ *Applied Categorical Data Analysis* 3(3+0)

Prasyarat : STS3410

Kursus ini merangkum asas bagi kaedah dan model yang digunakan untuk analisis and penyelesaian masalah berkaitan data berkategori dan data boleh bilang. Pentaabiran berkaitan analisis data berkategori dibincangkan.

This course covers the basic methods and models used to analyze and solve practical problems involving categorical and count data. Inference related to categorical data analysis is discussed.

STS4406 Siri Masa/ *Time Series* 3(3+0)

Prasyarat : STS3411

Kursus ini merangkum pengetahuan dan kefahaman data yang mempunyai kebergantungan dengan data sebelumnya. Aspek teori dan praktikal untuk menghasilkan model data siri masa pegun dibincangkan.

This course covers the knowledge and understanding of data that have dependencies with previous data. Theoretical and practical aspects of generating a stationary time series data model are discussed.

STS4407 Kaedah Interaktif Berkomputasi dalam Analisis Data/ *Interactive Computational Methods In Data Analysis* 3(3+0)

Prasyarat : STS3405

Kursus ini merangkum analisis data menggunakan pakej berstatistik terpilih. Penghasilan algoritma bagi analisis data interaktif turut dibincangkan.

This course covers data analysis using selected statistical packages. The development of algorithms for interactive data analysis is also discussed.

STS4408 Analisis Data Mandirian/ *Survival Data Analysis* 3(3+0)

Prasyarat : STS3405

Kursus ini memperkenalkan konsep dan pendekatan yang diperlukan untuk menganalisis data mandirian. Kaedah pentaabiran bagi pelbagai jenis data mandirian serta aplikasinya turut dibincangkan.

This course introduces the concepts and tools required for the analysis of survival data. Inferential methods for various types of survival data and their applications are also discussed.

STS4409 Ekonometrik Gunaan/ *Applied Econometrics* 3(3+0)

Prasyarat : STS3401

Kursus ini merangkum kaedah statistik yang sesuai bagi penyelesaian masalah dalam bidang ekonomi. Analisis regresi bagi model ekonometrik dan persamaan serentak juga dibincangkan.

This course covers appropriate statistical methods for solving problems in the field of economics. Regression analysis for econometric and simultaneous equation models are also discussed.

STS4410 Pembelajaran Mesin Dalam Statistik/ *Machine Learning in Statistics* 3(3+0)

Prasyarat : STS3411 dan STS3412

Kursus ini merangkum model dan algoritma pembelajaran mesin statistik untuk masalah praktikal. Peramalan untuk data raya dibincangkan.

This course covers statistical machine learning models and algorithms for practical problems. Forecasts for big data are discussed.

STS4411 Analisis Pengoptimuman Permukaan Sambutan/ *Analysis of Response Surface Optimization* 3(3+0)

Prasyarat : STS3403

Kursus ini merangkum konsep dalam reka bentuk dan alat pengoptimuman menggunakan kerangka permukaan tindak balas. Analisis pengoptimuman terpilih turut dibincangkan.

This course covers concepts in design and optimization tools using a response surface framework. Selected optimization analysis are also discussed.

STS4412 Analisis Terokaan Data/ *Exploratory Data Analysis* 3(3+0)

Prasyarat : STS3405

Kursus ini merangkum teknik visualisasi dan analisis terokaan data. Pengaturcaraan yang sesuai juga dibincangkan.

This course covers visualization technique and exploratory data analysis. Appropriate programming is also discussed.

STS4413 Statistik Ruang Gunaan/ *Applied Spatial Statistics* 3(3+0)

Prasyarat : STS3402

Kursus ini merangkum analisis data ruang. Analisis titik ruang, taburan komponen asas, taburan am dan taburan teritlak turut dibincangkan.

This course covers data analysis using selected statistical packages. The development of algorithms for interactive data analysis is also discussed.

STS4959 Disertasi Bachelo/ *Bachelor Dissertation* 6(0+6)

Prasyarat : Dengan kebenaran jabatan

Kursus ini merangkum sorotan kepustakaan, metodologi penyelidikan yang sesuai, pengumpulan dan analisis data, pentafsiran keputusan, perbincangan dan kesimpulan kajian saintifik serta pembentangan hasil penyelidikan.

This course covers literature review, appropriate research methodology, data collection and analysis, interpretation of results, discussion and conclusion of scientific studies and presentation of research output.